

Neg Cards

C1 - Econ

It's strong now.

Shipman 7-13 [Alan (Senior Lecturer in Economics @ The Open University), 07-13-2026, "Why the US economy stays strong even when its policies shock the rest of the world", Yahoo News, accessed 07-13-2026] prav

The **US economy** is continuing to **grow** faster and generate more new **jobs** than Europe. Annual national **income growth over the past five years has averaged 3.3%** in the US against 2.6% in the EU. In the first quarter of 2026, the EU's **GDP was** just 0.7% higher than a year before, while that of the US was up **2.6% on comparable measures**.¶ These figures **defy the widespread predictions that the US would lose its growth advantage** after its government imposed a global trade tariff regime in 2025 and, one year later, started a war with Iran. Economists see several factors behind the resilience of the US economy.¶ The US runs consistently wider budget deficits than the EU, UK or China. **By spending more than it collects in tax, the US government creates more income for the people it employs** and the businesses it buys from. This extra income in theory boosts demand in the economy, **pushing output growth higher and reducing unemployment**.¶ Most European governments also run budget deficits. The average budget deficit of EU countries in 2025, for example, was 3.1% of GDP. But the US deficit, at 5.8% of GDP that same year, is giving a much stronger stimulus.¶ The US also channels a higher proportion of its GDP into business investment and research and development than the EU. Europe was spending €270 billion (£230 billion) less than the US on innovation in 2021, with this spending concentrated on its century-old car industry rather than new technologies.¶ Since 2025, **AI has been the focus of US investment**. This has helped the US maintain its hold over global technology and digital platforms. Rapid uptake of AI across US industry has also widened the margin by which its labour productivity growth is outpacing Europe's. Output per hour in professional services has increased by over 18% since 2019 in the US compared to just 5% in the EU.¶ A man works at a computer in an office.¶ Uptake of AI across industry has also boosted labour productivity growth in the US. DC Studio / Shutterstock.¶ **Economy-wide productivity gains have allowed US real wages** (wages adjusted for inflation) **to edge higher** since 2019. This has **sustained consumer demand while also enabling the strong profit growth that has lifted US share prices to record levels**. In contrast, average real wages in the EU have barely grown over the past 20 years while corporate profits in Europe remain subdued.¶ The US technological lead could be dented by Donald Trump's immigration clampdown, which extends to skilled scientists and students. Research suggests annual GDP growth rates in the US could currently be as much as 0.8 percentage points lower than if net unauthorised immigration had stayed on its pre-2025 trend.¶ But the Trump administration and its tech-entrepreneur supporters also credit their success to more freedom to gamble with new ideas, while Europe regulates them more heavily and China tries to harness them for state control. Although the EU generates as many tech start-ups as the US, many relocate there when they start to expand.¶ Another factor explaining the resilience of the US economy is that American industry benefits from substantially lower energy costs than in Europe. The US produces more fossil fuels than Europe and taxes them less. It is also advancing fast with cheap renewable sources, despite the government's scepticism towards solar and wind.¶ Reliance on fossil fuels, and indifference to carbon emissions, may raise the US's long-term economic vulnerability. But for now they ensure a cost advantage that is allowing the US to regenerate its manufacturing and meet much of the global demand for data-based services such as e-commerce and generative AI.

Data centers are key.

Lichtenberg 25 Nick Lichtenberg [Nick Lichtenberg is Business Editor at Fortune, responsible for reporting and editing a mix of breaking news and digital features. This includes stories developed with the assistance of artificial intelligence, filed under Fortune Intelligence, as well as commentary submissions and book excerpts. Nick was formerly Fortune's executive editor of global news. Prior to Fortune, he was a deputy editor at Business Insider covering the economy, wealth, and real estate, and a breaking news editor at Bloomberg covering corporate news and equities markets.] October 7, 2025, Without data centers, GDP growth was 0.1% in the first half of 2025, Harvard economist says Fortune <https://fortune.com/2025/10/07/data-centers-gdp-growth-zero-first-half-2025-jason-furman-harvard-economist/> Accessed 06/29/26 nirm

U.S. GDP growth in the first half of 2025 was almost **entirely driven by investment in data centers** and information processing technology, according to Harvard economist Jason Furman. **Excluding** these **tech**nology-related categories, Furman calculated in a Sept. 27 post on X.com GDP **growth would have been** just 0.1% on an annualized basis, a near **standstill** that underlines the increasingly pivotal role of **high-tech infrastructure** in **shaping macroeconomic outcomes**. Furman's findings, shared online and echoed by financial analysts including Robert Armstrong of the Financial Times' Unhedged (the same writer who coined the term "TACO trade"), echo several months of observations on the remarkable **surge in data-center infrastructure**. In August, Renaissance Macro Research estimated, to date in 2025, the dollar value **contributed to GDP growth** by AI data-center buildout had **surpassed U.S. consumer spending** for the first time ever. That's remarkable considering consumer spending is two-thirds of GDP. Technically, as Furman notes, investment in information-processing equipment and software was only 4% of U.S. GDP for the first half of 2025, yet it also accounted for fully **92% of GDP growth** over that period. Furman added it's probably not the case the U.S. economy would have recorded almost no expansion at all absent this buildout, reasoning that "absent the AI boom we would probably have lower interest rates [and] electricity prices, thus some additional growth in other sectors. In very rough terms that could maybe make up about half of what we got from the AI boom." But still, it's big. Tech giants such as Microsoft, Google, Amazon, Meta, and Nvidia have poured tens of billions of dollars into building and upgrading data centers, responding to explosive demand for artificial intelligence and large language models that require massive computing resources. Lisa Shallet, chief investment officer for Morgan Stanley Wealth Management, flagged on Sept. 29 that spending was truly massive among the so-called "**hyperscalers**" who are **striving** for huge computing, storage and networking capacity. **"In recent years, hyperscaler capex on data center and related items has risen fourfold and is nearing \$400 billion annually,"** she wrote. **"The speed of growth and size of the investment are skewing its aggregate economic impact, with the top 10 spenders accounting for nearly a third of all spending ... For perspective, it's estimated that data center-linked spending is adding roughly 100 basis points to U.S. real GDP growth."** The 'mystery' of the economy This **surge in technology-led growth comes against a backdrop of wider economic sluggishness** and paradoxically strong GDP growth. **Job creation has slowed**, raising concerns that, **absent technology investment**, the **U.S.** economy could have **slipped into recession**. **Other sectors—from manufacturing and real estate to retail and services—contributed little or even detracted from overall output in the first half of 2025.** And yet, as Apollo Global Management Chief Economist Torsten Słøk has noted, the GDP figures speak of a (statistically) strong economy. **"The consensus has been wrong** since January," Słøk said in a note circulated to clients in early October, adding the average of economists' forecasts has said the U.S. economy would slow down for nine months consecutively. "But the reality is that it has simply not happened ... We in the economics profession need to look ourselves in the mirror." Furman's analysis adds to the snarky and accurate observation by Rusty Foster of Today in Tabs who quipped: **"Our economy might just be three AI data centers in a trench coat"**—an allusion to the data-center buildout boom and to the cartoon trope/sight gag of several young boys teaming up to disguise themselves as an adult.

The AFF gets rid of it.

Gerstner 26 [Brad Gerstner [Bradley Thomas Gerstner is an American entrepreneur, investor, venture capitalist, hedge fund manager, and podcaster. He is the founder, chairman, and CEO of Altimeter Capital. Gerstner appeared on the 2022 Forbes Midas List after his firm's] May 29, 2026 Where Brad Gerstner Is Investing Billions PodWave TBPN <https://pod.wave.co/podcast/tbpn/where-brad-gerstner-is-investing-billions> Accessed 06/29/26 prav

And lest we be overconfident in Silicon Valley, let's remember the activists. A small group of activists shut down supersonic technology, and a small group of activists shut down all nuclear clean energy in this country. Okay. We have 100 fission reactors being built in China. We have one in the United States. It's a disaster that happened. And so we can't take for granted that the kooks who are calling for data center moratoriums. Right. Which just think about this for a second. **All of our GDP growth is coming from the fact that we are building data centers and driving AI and driving productivity improvements in the economy. A data center moratorium would thrust us straight into a recession** and high unemployment. Secondly, **it would cede the entire global game to China. Like overnight we would lose to China in the global air race, which is not just about AI, but it's about economic security, it's about jobs and it's about national security.** So it literally is insane that we would do this. I can't even believe there are people talking about it. However, why are they talking about it? Because

people are concerned. Local communities are concerned. I just got back from celebrating my mother's 90th in rural Indiana, you know, over the Memorial Day weekend. And what I'll tell you.

Decline causes great power war.

Jung 24 [Sung Chul Jung (Associate Professor in the Department of Political Science and Diplomacy), 2024, Myongji University, Journal of Peace Research, 61(2) pg. 180-196, "Economic slowdowns and international conflict," <https://ideas.repec.org/a/sae/joupea/v61y2024i2p180-196.html>, accessible @ https://drive.google.com/file/d/1mlanhmQ_rMFdHxOs4-Svbi8JZeUONg_c/view?usp=sharing, accessed 10-11-25] **we do not endorse problematic language**

In sum, this statistical analysis shows that an economic slowdown increases the likelihood of initiating military conflict. Specifically, economically troubled autocracies act more aggressively toward democracies whose economies depend on them. Political unrest also increases conflict initiation, as many studies of diversionary conflict have argued. However, autocracies suffering from political troubles do not seem to target economically dependent democracies. Democracies with smaller economies seem to be attractive targets for economically troubled autocracies whose leaders may blame the politically different partner's 'unfair' trade and currency practices and domestic policies for the deteriorating economy. The autocracies will present their military decisions as inevitable to correct international wrongs and reboot the national economy. Meanwhile, leaders facing political unrest may seek diversionary targets, initiating a conflict through which they can mobilize their support base (and marginalize opposition groups) by exploiting ideological, religious, and/or ethnic identities of us' and 'them'. Future studies should seek to unravel the complex relationship between types and levels of domestic unrest and types and combinations of political regimes, in either the presence or absence of diversionary conflict. The findings of this study suggest that an attractive target is not predetermined but is closely related to the domestic and international situation confronted by leaders who are battling for their political survival. ¶ Summary and implications ¶ Since the 2000s, many pundits and scholars have attributed two rising powers' instances of aggression to their domestic problems. Russia's invasion of Ukraine and annexation of Crimea and its deployment of troops to Syria, along with China's assertive actions in the South China Sea, have been seen as aggressions arising out of domestic weakness (Kaplan, 2016). Some have attributed China's recent provocative actions against India, Japan, and Taiwan to its struggling economy in the wake of the pandemic (Myers, 2020). In the post-Cold War period, these two great powers, often regarded as revisionist powers, have regained their national capabilities to some extent and have sought higher status in international politics (Larson & Shevchenko, 2010). What made them more aggressive in the 2010s? One promising explanation is their decreasing economic growth rates. Due to their abundance of natural and human resources, the two former communist countries had maintained their increased status relative to their neighboring states since the 2000s or even earlier. However, falling energy prices and the reform-incapable structures of their economies retarded their further growth and increased the burden on their authoritarian leaders, who also had to manage domestic calls for political freedom and maintain the legitimacy of their rule. According to this article's causal explanation, Russia and China became more revisionist actors in the 2010s at least partly because their economies went downhill, and their aggressions were directed at smaller and more vulnerable economies. ¶ A slowing economy and economic dependence have both individual and interactive effects on military conflict in international politics. Slow growth rates contribute positively to conflict initiation, especially when an autocracy suffering from a slowing economy targets a democracy whose economy is dependent on the autocracy. Whereas economic interdependence has been regarded by many liberal scholars as a reason for international cooperation, this study shows why and how asymmetric interdependence contributes to interstate conflict by articulating its interaction with an economic slowdown in a dyadic relationship. Both political unrest and economic problems have destabilizing effects on international politics, but their effects differ when economically dependent democracies are the target. Future research should compare different types of domestic unrest and their effects on international relations with sophisticated logic and strong evidence. ¶ This analysis shows that economic growth has two distinct effects. Slower growth makes one state weaker than others and its leaders more vulnerable to domestic challenges. A state's economic slowdown not only causes a change in the balance of economic power so as to favor its competitor but also increases the troubled state's foreign aggression, which can harm its adversary. Therefore, it is uncertain whether one state's economic slowdown benefits its rival's security. An autocracy's slowing economy can be a warning, rather than good news, to its democratic opponents if the latter maintain economic relationships with the former. In this regard, economic sanctions may provoke the target leader's diversionary tactic of blaming its poor economy on the sanctioner. We can see this pattern in Tokyo's surprise attack on Pearl Harbor at a time of US sanctions against Japan, which had continued since the late 1930s. ¶ The current global economic outlook is uncertain in the wake of COVID-19. Although we do not yet know the pandemic's long-term economic consequences, all states will likely struggle with slower growth and the erosion of democracy and will engage in more protectionist policies to reboot their economies and save major industries. In this context, we should be especially concerned about the risk that troubled economies pose to international security. As this study's findings suggest, slower growth causes domestic problems, which often lead to the diversionary

use of force against foreign states. If we cannot prevent struggling leaders from using diversionary tactics, **worsening economies** in authoritarian states **will likely drive such states into a more** confrontational or **aggressive stance toward their economic partners**, if not toward political and ideological partners. We can all hope world leaders will realize that this is a time to beat a global pandemic and an economic recession, not to beat up on a foreign state.

Extinction

Clare 25 [Stephen Clare (former research fellow @ the Forethought Foundation) 03-xx-2025, “Great power war”, 80,000 Hours, <https://80000hours.org/problem-profiles/great-power-conflict/>, accessed 08-12-25]

A modern great power war **could see nuclear weapons, bioweapons, autonomous weapons, and other destructive new technologies deployed on an unprecedented scale.** **It would probably be the most destructive event in history, shattering our world. It could even threaten us with**

extinction. We’ve come perilously close to just this kind of catastrophe before. On October 27, 1962 — near the peak of the Cuban Missile Crisis — an American U-2 reconnaissance plane set out on a routine mission to the Arctic to collect data on Soviet nuclear tests. But, while flying near the North Pole, with the stars obscured by the northern lights, the pilot made a navigation error and strayed into Soviet airspace.¹ Soviet commanders sent fighter jets to intercept the American plane. The jets were picked up by American radar operators and nuclear-armed F-102 fighters took off to protect the U-2. Fortunately, the reconnaissance pilot realised his error with enough time to correct course before the Soviet and American fighters met. But the intrusion enraged Soviet Premier Nikita Khrushchev, who was already on high alert amidst the crisis in Cuba. “What is this, a provocation?” Khrushchev wrote to US President John F. Kennedy. “One of your planes violates our frontier during this anxious time when everything has been put into combat readiness.” If the U-2’s path had strayed further west, or the Soviet fighters had been fast enough to intercept it, this incident could have played out quite differently. Both the United States and the USSR had thousands of nuclear missiles ready to fire. Instead of a nearly-forgotten anecdote, the U-2 incident could have been a trigger for war, like the assassination of Franz Ferdinand. Competition among the world’s most powerful countries shapes our world today. And whether it’s through future incidents like the lost U-2, or something else entirely, it’s plausible that it could escalate and lead to a major, devastating war. Is there anything you can do to help avoid such a terrible outcome? It is, of course, difficult to imagine how any one individual can hope to influence such world-historical events. Even the most powerful world leaders often fail to predict the global consequences of their decisions. But I think the likelihood and severity of great power war makes this among the most pressing problems of our time — and that some solutions could be impactful enough that working on them may be one of the highest-impact things to do with your career. By taking action, I think we can create a future where the threat of great power war is a distant memory rather than an ever-present danger. Summary **Economic growth and technological progress have bolstered the arsenals of the world’s most powerful countries.** That means the next war between them could be far worse than World War II, the deadliest conflict humanity has yet experienced. Could such a war actually occur? We can’t rule out the possibility. **Technical accidents or diplomatic misunderstandings could spark a conflict that quickly escalates. Or international tension could cause leaders to decide they’re better off fighting than negotiating.**

It seems hard to make progress on this problem. It’s also less neglected than some of the problems that we think are most pressing. There are certain issues, like making nuclear weapons or military artificial intelligence systems safer, which seem promising — although it may be more impactful to work on reducing risks from AI, bioweapons or nuclear weapons directly. You might also be able to reduce the chances of misunderstandings and miscalculations by developing expertise in one of the most important bilateral relationships (such as that between the United States and China). Finally, by making conflict less likely, reducing competitive pressures on the development of dangerous technology, and improving international cooperation, you might be helping to reduce other risks, like the chance of future pandemics. Our overall view: Recommended. Working on this issue seems to be among the best ways of improving the long-term future we know of, but all else equal, we think it’s less pressing than our highest priority areas (primarily because it seems less neglected and harder to solve). **Suite** There’s a significant chance that a new great power war occurs this century. Although the world’s most powerful countries haven’t fought directly since World War II, war has been a constant throughout human history. There have been numerous close calls, and several issues could cause diplomatic disputes in the years to come. These considerations, along with forecasts and statistical models, lead me to think there’s about a one-in-three chance that a new great power war breaks out in roughly the next 30 years. Few wars cause more than a million casualties and the next great power war would probably be smaller than that. However, there’s some chance it could escalate massively. Today the great powers have much larger economies, more powerful weapons, and bigger military budgets than they did in the past. An all-out war could kill far more people than even World War II, the worst war we’ve yet experienced. It could become an existentially threatening war — one that could cause human extinction or significantly damage the prospects of the long-term future. It’s very difficult to say, but my best current guess is that the chance of an existential catastrophe due to war in the next century is somewhere between 0.05% and 2%. Neglecters There is a lot less neglected than some of our other top problems. There are thousands of people in governments, think tanks, and universities already working on this problem. But some solutions or approaches remain neglected. One particularly promising approach is to develop expertise at the intersection of international conflict and another of our top problems. Experts who understand both geopolitical dynamics and risks from advanced artificial intelligence, for example, are sorely needed. Solvability Reducing the risk of great power war seems very difficult. But there are specific technical problems that can be solved to make weapons systems safer or less likely to trigger catastrophic outcomes. And in the best case, working on this problem can have a leverage effect, making the development of several dangerous technologies safer by improving international cooperation and making them less likely to be deployed in war. At the end of this profile, I suggest five issues which I’d be particularly excited to see people work on. These are: Developing expertise in the riskiest bilateral relationships. Learning how to manage international crises quickly and effectively and ensuring the systems to do so are properly maintained. Being research to improve particularly important foreign policies, like strategies for sanctions and deterrence. Improving how nuclear weapons and other weapons of mass destruction are governed at the international level. Improving how such weapons are controlled at the national level. Why might preventing great power war be an especially pressing problem? A modern great power war — an all-out conflict between the world’s most powerful countries — could be the worst thing to ever happen to humanity. Historically, such wars have been exceptionally destructive. Sixty-one million people died in World War II, likely the deadliest catastrophic event humanity has experienced so far. Since World War II, the global population and world economy have continued to grow, nuclear weapons have proliferated, and military technology has continued to advance. This means the next world war could be even worse, just as World War II was much deadlier than World War I. It’s not guaranteed that such a war will break out. And if it does, it may not escalate to such a terrible extent. But the chance can’t be ignored. In fact, there are reasons to think that the odds of World War II breaking out this century are worryingly high. A modern great power war would be devastating for people alive today, but its effects could also persist long into the future. That’s because there is a substantial chance that this century proves to be particularly important. Technologies with the potential to cause a global catastrophe or radically reshape society are likely to be invented. How we choose to develop and deploy them could impact huge numbers of our descendants. And these choices would be affected by the outcomes of a major war. To be more specific, there are three main ways great power conflict could affect the long-term future. High international tension could increase other risks. Great power tensions could make the world more dangerous even if they don’t lead to war. During the Cold War, for example, the United States and the USSR never came into direct conflict but invested in bioweapons research and built up nuclear arsenals. This dynamic could return, with tension between great powers fueling races to develop and build new weapons, raising the risk of a disaster even before shots are fired. War could cause an existential catastrophe. If war does break out, it could escalate dramatically, with modern weapons (nuclear weapons, bioweapons, autonomous weapons, or other future technologies) deployed at unprecedented scale. The resulting destruction could irreparably damage humanity’s prospects. War could reshape international institutions and power balances. While such a catastrophic war is possible, it seems extremely unlikely. But even a less deadly war, such as another conflict on the scale of World War II, could have very long-lasting effects. For example, it could reshape international institutions and the global balance of power. In a postwar century, different institutional arrangements and geopolitical balances could cause humanity to follow different long-term trajectories. The rest of this profile explores exactly how pressing a problem great power conflict is. In summary: Great power tensions have become more tense. More. Partly as a result, a war is more likely than you might think. It’s reasonable to put the probability of such a conflict in the coming decades somewhere between 10% and 50%. More. If war breaks out, it would probably be hard to control escalation. The chance that it would become large enough to be an existential risk cannot be dismissed. More. This makes great power war one of the biggest threats our species currently faces. More. It seems hard to make progress on solving such a difficult problem (more) — but there are many things you can try if you want to help (more). International tension has risen and makes other problems worse. Imagine we had a thermometer-like device which, instead of measuring temperature, measured the level of international tension. This ‘tension metre’ would max out during periods of the all-out global war, like World War II. And it would be relatively low when the great powers were peaceful and cooperative. For much of the post-Napoleonic 1800s, for example, the powerful European nations instituted the Concert of Europe and mostly upheld a continental peace. The years following the fall of the USSR also seem like a time of relative calm, when the tension metre would have been quite low. How much more worried would you be about the coming decades if you knew the tension metre would be very high than if you knew it would be low? Probably quite a lot. In the worst case, of course, the great powers could come into direct conflict. But even if it doesn’t lead to war, a high level of tension between great powers could accelerate the development of new strategic technologies, make it harder to solve global problems like climate change, and undermine international institutions. During the Cold War, for instance, the United States and USSR avoided coming into direct conflict. But the tension metre would still have been pretty high. This led to some dangerous events. A nuclear arms race. The number of nuclear warheads in the world grew from just 300 in 1950 to over 64,000 in 1986. The development of new bioweapons. Despite signing the Biological Weapons Convention in 1972, the search for military advantages motivated Soviet decision makers to continue investing in bioweapon development for decades. Although never used in combat, biological agents were accidentally released from research facilities, resulting in dozens of deaths and threatening to cause a pandemic. Nuclear close calls. Military accidents and false alarms happened regularly, and top decision makers were more likely to interpret these events hostilely when tensions were high. On several occasions it seems the decision about whether or not to start a nuclear war came down to individuals acting under stress and with limited tools. This makes international tension an existential risk factor. It’s connected to a number of other problems, which means reducing the level of international tension would lower the total amount of existential risk we face. The level of tension today. Recently, international tension seems to have once again been rising. To highlight some of the most salient examples: China-United States relations have deteriorated, leading to harsh diplomatic rhetoric and protectionist trade policies that aim to reduce the countries’ economic interdependence. Russia’s invasion of Ukraine has killed about a hundred thousand people so far, raised the risk of nuclear war, and sent United States-Russia relations to their lowest point since the Cold War. Chinese and Indian soldiers fought deadly skirmishes along their countries’ disputed border in 2020–21. These dynamics raise an important question: how much more dangerous is the world given this higher tension than it would be in a world of low tension? I think the answer is quite a bit more dangerous — for several reasons. First, international tension seems likely to make technological progress more dangerous. There’s a good chance that, in the coming decades, humanity will make some major technological breakthroughs. We’ve discussed, for example, why one might worry about the effects of advanced artificial intelligence systems or biotechnology. The level of tension could strongly affect how these technologies are developed and governed. Tense relations could, for example, cause countries to neglect safety concerns in order to develop technology faster. Second, great power relations will strongly influence how nations do, or do not, cooperate to solve other global collective action problems. For example, in 2022, China withdrew from bilateral negotiations with the United States over climate action in protest of what it perceived as American diplomatic aggression in Taiwan. That same year, efforts to strengthen the Biological Weapons Convention were reportedly hampered by the Russian delegation after their country’s invasion of Ukraine raised tensions with the United States and other weapons countries. And third, if relations deteriorate severely, the great powers could fight a war. How likely is a war? Wars are destructive and risky for all countries involved. Modern weapons, especially nuclear warheads, make starting a great power war today seem like a suicidal undertaking. But factors like the prevalence of war throughout history, the chance that baden make mistakes, conflicting ideologies, and commitment problems, make me think that conflict could break out anyway. On balance, I think such an event is somewhat unlikely but hardly unthinkable. To quantify this: I put the chance we experience some kind of war between great powers before 2050 at about one-in-three. War has occurred regularly in the past. One reason to think a war is quite likely is that such conflicts have been so common in the past. Over the past 350 years, about two great power wars have occurred per century. Nearly, this would mean that every year there’s a 2% chance such a war occurs, implying the chance of experiencing at least one great power war over the next 80 years — roughly until the end of the century — is about 80%. This is a very simple model. In reality, the risk is not constant over time and independent across years. But it shows that if bad trends simply continue, the outcome is likely to be very bad. Has great power war become less likely? One of the most important criticisms of this model is that it assumes the risk is constant over time. Some researchers have argued instead that, especially since the end of World War II, major conflicts have become much less likely due to nuclear deterrence. Nuclear weapons are so powerful and destructive that it’s just too costly for nuclear-armed countries to start wars against each other. Democratization. Democracies have almost never gone to war against each other, perhaps because democracies are more interconnected and their leaders are under more public pressure to peacefully resolve disputes with each

other 11 The proportion of countries that are democratic has increased from under 10% in 1945 to about 50% now. Strong economic growth and global trade: Global economic growth accelerated following World War II and the value of global exports grew by a factor of almost 30 between 1950 and 2014. Since war disrupts economies and international trade, strong growth raises the costs of fighting. The spread of international institutions: The United Nations General Assembly and Security Council promote diplomatic dialogue and facilitate coordination to punish transgressors. It is true that we are living through an unusually long period of great power peace. It's been about 80 years since World War I. We just saw that a simple model using the historical frequency of great power wars suggests there was only a 20% chance of going that long without at least one more war breaking out. This is some evidence in favor of the idea that we have become significantly less competitive. At the same time, we shouldn't feel too optimistic. The numerous close calls during the Cold War suggest we were somewhat lucky to avoid a major war in that time. And a 20% chance of observing 80 years of peace is not that low. Structural changes might have dramatically reduced the likelihood of war. Or perhaps we're just being lucky. It could even be that technological advances have made war less likely to break out, but more deadly when it occurs, leaving the overall effect on the level of risk ambiguous. It just hasn't been long enough to support a decisive view. So while the recent historical trend is somewhat encouraging, we don't have nearly enough data to be confident that great power war is a thing of the past. To better predict the likelihood of future conflict, we should also consider distinctive features of our modern world. One might think that a modern great power war would simply be so destructive that no state leader would ever choose to start one. And so many researchers do think that the destruction such a war would wreak globally makes it less likely to occur. But it would be hard to find anyone who claims this dynamic has driven the risk to zero. First, a war could be started by accident. Second, sometimes even prudent leaders may struggle to avoid a slide towards war. We could blunder into war. An accidental war can occur if one side mistakes some event as an aggressive action by an adversary. This happened several times during the Cold War. The earlier example of the supposed American reconnaissance plane shows how routine military exercises carry some escalation risk. Similarly, throughout history, nervous pilots and captains have caused serious incidents by attacking civilian planes and ships. 17 nuclear weapons allow for massive retaliatory strikes to be launched quickly – potentially too quickly to allow for such situations to be explained and de-escalated. It is perhaps more likely, though, that an accidental war could be triggered by a technological malfunction. Faulty computers and satellites have previously triggered nuclear close calls. As monitoring systems have become more reliable, the rate at which such accidents have occurred has been going down. But it would be overconfident to think that technological malfunctions have become impossible. Future technological changes will likely raise new challenges for nuclear weapon control. There may be pressure to integrate artificial intelligence systems into nuclear command and control to allow for faster data processing and decision making. And AI systems are known to behave unexpectedly when deployed in new environments. New technologies will also create new accident risks of their own, even if they're not connected to nuclear weapon systems. Although these risks are hard to predict, they seem significant. It's say more about how such technologies – including AI, nuclear, biological, and autonomous weapons – are likely to increase war risks later. Leaders could choose war. All that said, most wars have not started by accident. If another great power war does break out in the coming decades, it is more likely to be an intentional decision made by a national leader. Explaining why someone might make such a costly, unpredictable, and risky decision has been called "the central puzzle about war." It has motivated researchers to search for "rational" explanations for war. In his 2022 book Why We Fight, for example, economist Chris Blattman proposes five basic explanations: uncharted interests, intangible incentives, uncertainty, commitment problems, and misperceptions. This section discusses how great power tensions may escalate to war in the next few decades. It focuses on three potential conflicts in particular: war between the US and China, between the US and Russia, and between China and India. These are discussed because each of these countries are among the world's largest economies and military spenders, and seem particularly likely to fight. At the end, I briefly touch on other potential large conflicts. United States-China The most worrying possibility is war between the United States and China. They are easily the world's largest economies. They spend by far the most on their militaries. Their diplomatic relations are tense and have recently worsened. And their relationship has several of the characteristics that Blattman identifies as causes of war. At the core of the United States-China relationship is a commitment problem. China's economy is growing faster than the United States'. By some metrics, it is already larger. If its differential growth continues, the gap will continue to widen between it and the United States. While economic power is not the sole determinant of military power, it is a key factor. The United States and China may be able to strike a fair deal today, but as China continues to grow faster, that deal may come to seem unbalanced. Historically, such commitment problems seem to have made these kinds of transition periods particularly dangerous. In practice, the United States and China may find it hard to agree on rules to guide their interactions, such as how to run international institutions or govern areas of the world where their interests overlap. The most obvious issue which could tip the United States-China relationship from tension into war is a conflict over Taiwan. Taiwan's location and technology industries are valuable for both great powers. This issue is further complicated by intangible incentives. For the United States, it is also a conflict over democratic ideals and the United States' reputation for defending its allies. For China, it is also a conflict about territorial integrity and advancing what are seen as past injustices. Ask, forecasts suggest that while a conflict is certainly possible, it is far from inevitable. As of 8 June 2023, our aggregated forecast gives a 17% chance of a United States-China war breaking out before 2035. A related aggregated forecast of the chance that at least 100 deaths occur in conflict between China and Taiwan by 2050 gives it, as of 8 June 2023, a much higher 68% chance of occurring. United States-Russia Russia is the United States' other major geopolitical rival. Unlike China, Russia is not a rival in economic terms: even after adjusting for purchasing power, its economy is only about one-fifth the size of the United States'. However, Russia devotes a substantial fraction of its economy to its military. Crucially, it has the world's largest nuclear arsenal. And Russian leadership has shown a willingness to project power beyond their country's borders. Top five countries by estimated military spending, 2021. Source: SIPRI. Russia's 2022 invasion of Ukraine demonstrated the dangers of renewed rivalry between Russia and the United States-led West. The war has already been highly destructive: the largest war in Europe since World War I, with hundreds of thousands of casualties already and no end to the conflict in sight. And it could get much worse. Most notably, Russian officials have repeatedly urged to rule out the use of nuclear weapons. Unchecked interests and intangible incentives are again at play here. Vladimir Putin leads a highly-centralized government. He has spoken about how his desire to rebuild Russia's reputation played in his decision to invade Ukraine. Given their ideological differences and history of rivalry, it is reasonable to expect that the United States and Russia will continue to experience dangerous disagreements in the future. As of 8 June 2023, an aggregated forecast gives a 20% chance that the United States and Russia will fight a war involving at least 1,000 battle deaths before 2035. China-India India is already the world's third largest economy. If national growth rates remain roughly constant, the size of the Indian economy will surpass that of the United States' sometime this century. India also has nuclear weapons and is already the world's third-largest military spender (behind at a much lower level than China or the United States). One reason to worry that China and India could fight a war is that they already dispute territory along their border. Countries that share a border, especially when it is disputed, are more likely to go to war than those that do not. By one count, 88% of the wars that occurred between 1816 and 1982 began as wars between neighbors. In fact, China and India already fought a border but violent border war in 1962. Deadly skirmishes have continued since, resulting in deaths as recently as 2020. Forecasts agree that a China-India conflict seems relatively (though not absolutely) likely. An aggregated forecast gives a 18% chance of war before 2035. Other dangerous conflicts These three conflicts – United States-China, United States-Russia, and China-India – are not the only possible great power wars that could occur. Other potential conflicts could also pose existential risk, either because they drive dangerous arms races or see widespread deployment of dangerous weapons. We should keep in mind India-Pakistan as a particularly likely conflict between nuclear-armed states and China-Russia as a potential, though unlikely, conflict between great powers with a disputed border and history of war. Plus, new great powers may emerge or current great powers may fade in the years to come. While I think we should prioritize the three potential conflicts I've highlighted above, the future is highly uncertain. We should monitor geopolitical changes and be open to changing our priorities in the future. Overall predictions Below is a table listing relevant predictions from the forecasting platform Metaculus, including the number of predictions made, as of 10 March 2023. Note the different timeframes and resolution criteria for each question; they may not be intuitively comparable. I have previously independently estimated the likelihood of seeing a World War II-like conflict this century. My calculation first adjusts historical base rates to allow for the possibility that major wars have become somewhat less likely, and uses the adjusted base rate to calculate the probability of seeing a war between now and 2120. This method gives a 45% chance of seeing a major great power war in the next 77 years. If the probability is constant over time then the cumulative probability between now and 2050 would be 22%. This is aligned with the Metaculus predictions above. We can also ask experts what they think. Unfortunately, there are surprisingly few predictions about the likelihood of major conflict. One survey was conducted by the Project for the Study of the 21st Century. The numbers were relatively aligned with the Metaculus forecasts, though slightly more pessimistic. However, it seems a mistake to put too much stock in this survey (see footnote). We now have at least a rough sense of a great power war's probability, but how bad could it get if it occurred? In a new great power war could be devastating. At the time, the mechanized slaughter of World War I was a shocking step-change in the potential severity of warfare. But its severity was surpassed just 20 years later by the outbreak of World War II, which killed more than twice as many people. A modern great power war could be even worse. How bad have wars been in the past? The graph below shows how common wars of various sizes are, according to the Correlates of War's Interstate War dataset. This axis here represents war size in terms of the logarithm of the number of battle deaths. The y-axis shows the rate in the dataset that are at least that large. Using logarithms means that each step to the right in the graph represents a war not one unit larger, but 10 times larger. And each step up represents a war that is not one unit more likely, but 10 times more likely. Other the graph shows it is that wars have a heavy tail. Most wars remain relatively small. But a few escalate greatly and become much worse than average. Of the 95 wars in the latest version of the database, the median battle death count is 8,000. But the heavy tail means the average is 334,000 battle deaths. And the worst war, World War I, had almost 20 million battle deaths. The number of battle deaths is only one way to measure the badness of wars. We could also consider the proportion of the population of the countries involved who were killed in battle. By this measure, the worst war since 1816 was not World War I. Instead, it's the Paraguayan War of 1864-70. In that war, 30 soldiers died for every 1,000 citizens of the countries involved. It's even worse if we also consider civilian deaths, while estimates are very uncertain, it's plausible that about half of the men in Paraguay, or around a quarter of the entire population, was killed. What if instead we compared wars by the proportion of the global population killed? World War I is again the worst conflict since 1816 in this measure, having killed about 3% of the global population. Going further back in time, though, we can find worse wars. Ghengis Khan's conquests likely killed about 5-15% of people in the world at the time. The heavy tail means that some wars will be shockingly large. The scale of World War I and World War II took people by surprise, including the leaders who initiated it. It's also hard to know exactly how big wars could get. We haven't seen many really large wars. So while we know there's a heavy tail of potential outcomes, we don't know what that tail looks like. That said, there are a few reasons to think that wars much worse than World War I are possible. We're institutionally unlikely to have brushed up against the end of the tail, even if the tail has an upper bound. Other wars have been deadlier on a per-capita basis. So unless wars involving countries with larger populations are systematically less intense, we should expect to see more

intense wars involving as many people as World War I. Economic growth and technological progress are continually increasing humanity's war-making capacity. This means that, once a war has started, we're at greater risk of extremely bad outcomes than we were in the past. So how bad could it get?

How bad could a modern great power war be?

Over time, two related factors have greatly increased humanity's capacity to make war. First, scientific progress has led to the invention of more powerful weapons and improved military efficiency. Second, economic growth has allowed states to build larger armies and arsenals. Since World War II, the world economy has grown by a factor of more than 10 in real terms; the number of nuclear weapons in the world has grown from basically none to more than 9,000, and we've invented drones, missiles, satellites, and advanced planes, ships, and submarines. Ghengis Khan's conquests killed about 10% of the world, but this took place over the course of two decades. Today that proportion may be killed in a matter of hours. First, nuclear weapons could be used. Today there are around 10,000 nuclear warheads globally. At the peak of nuclear competition between the United States and the USSR, though, there were 64,000. If arms control agreements break down and competition resurges among two or even three great powers, nuclear arsenals could expand. In fact, China's arsenal is very likely to grow – though by how much remains uncertain. Many of the nuclear weapons in the arsenals of the great powers today are at least 10 times more powerful than the atomic bombs used in World War II. Should these weapons be used, the consequences would be catastrophic. By any measure, such a war would be by far the most destructive, dangerous event in human history, with the potential to cause billions of deaths. The probability that it would, on its own, lead to humanity's extinction or unrecoverable collapse, is contested. But there seems to be some possibility – whether through a famine caused by nuclear winter, or by reducing humanity's resilience enough that something else, like a catastrophic pandemic, would be far more likely to reach extinction-levels (read more in our problem profile on nuclear war). Nuclear weapons are complemented and amplified by a variety of other modern military technologies, including improved missiles, planes, submarines, and satellites. They are also not the only military technology with the potential to cause a global catastrophe – bioweapons, too, have the potential to cause massive harm through accidents or unexpected effects. What's more, humanity's war-making capacity seems poised to further increase in the coming years due to technological advances and economic growth. Technological progress could make it cheaper and easier for more states to develop weapons of mass destruction. In some cases, political and economic barriers will remain significant. Nuclear weapons are very expensive to develop and there exists a strong international taboo against their proliferation. In other cases, though, the hurdles to developing extremely powerful weapons may prove lower. Improvements in biotechnology will probably make it cheaper to develop bioweapons. Such weapons may provide the deterrent effect of nuclear weapons at a much lower price. They also seem harder to monitor from abroad, making it more difficult to limit their proliferation. And they could spark a global biological catastrophe, like a major – possibly existentially catastrophic – pandemic. Artificial intelligence systems are also likely to become cheaper as well as more powerful. It is not hard to imagine important military implications of this technology. For example, AI systems could control large groups of lethal autonomous weapons (though the timeline on which such applications will be developed is unclear). They may increase the pace at which war is waged, enabling rapid escalation outside human control. And AI systems could speed up the development of other dangerous new technologies. Finally, we may have to deal with the invention of other weapons which we can't currently predict. The feasibility and danger of nuclear weapons was unclear to many military strategists and scientists until they were first tested. We could similarly experience the invention of destabilising new weapons in our lifetime. What these technologies have in common is the potential to quickly kill

huge numbers of people: A nuclear war could kill tens of millions within hours, and many more in the following days and months. A runaway bioweapon could prove very difficult to stop. Future autonomous systems could act with lightning speed, even taking humans out of the decision-making loop entirely. Faster wars leave less time for humans to intervene, negotiate, and find a resolution that limits the damage. How likely is war to damage the long-run future? When a war begins, leaders often promise a quick, limited conflict. But escalation proves hard to predict ahead of time (perhaps because people are scope-insensitive, or because escalation depends on idiosyncratic decisions). This raises the possibility of enormous wars that threaten all of humanity.

C2 - Military

New military data center construction is key

Dickstein 26 [Corey (reporter @ Stars and Stripes), "Commercial 'hyperscale data centers' coming to Texas, Utah Army bases," Stars and Stripes, 03-26-2026, <https://www.stripes.com/branches/army/2026-03-26/army-bases-ai-data-centers-21188343.html>, DOA 07-06-2026] prav

The U.S. Army selected two commercial companies to build AI data centers on Fort Bliss, Texas, and Dugway Proving Ground, Utah, the service announced Thursday. The Army will enter exclusive negotiations with Carlyle, an investment firm, and CyrusOne, a data center developer, to build "hyperscale data centers" on the Army posts as the service seeks to expand its use of AI in combat and other military applications. The agreements would allow the companies to build, operate and maintain the data centers on land leased from the Army, according to a service news release. "AI is a **strategic asset** for the Army," Army Secretary Dan Driscoll said in a statement. "It is a force multiplier, supports future transformation and requirements, keeps the Army ahead of our adversaries, and generates resiliency across the force." These data centers are a critical resource to support that strategic imperative." Carlyle and CyrusOne would be responsible for financing the projects, and taxpayers would not see an upfront cost to their development, Army officials said. The Army Corps of Engineers will conduct final lease negotiations and oversee the projects, including conducting environmental reviews, according to the Army. The agreements come after a 2025 executive order from President Donald Trump that sought to make available underutilized federal land for commercial data center projects. Carlyle was selected to build on about 1,384 acres at Fort Bliss, according to the Army. That project is expected to be functional in 2027. CyrusOne, a company held by investment giants KKR and BlackRock, was selected to build on about approximately 1,201 acres on Dugway Proving Ground. That project is expected to be completed around 2029. Each project is estimated to cost about \$2 billion, according to the Financial Times. The Army in January solicited bids for leases to build commercial data centers on Fort Bliss, Dugway Proving Ground, Fort Hood in Texas and Fort Bragg in North Carolina. Army officials have said working hand-in-hand with the private sector is necessary to leverage ever-advancing technology into soldiers' hands. The service is also seeking private companies to build nuclear microreactors on some of its installations, which officials have said would bolster power needs for the service, including expanded AI capabilities. "By partnering with the private sector to develop cutting-edge data centers on our installations, we are bolstering our national security, driving technological innovation, and building a more resilient and modern Army," David R. Fitzgerald, the deputy undersecretary of the Army, said in a statement. "Our new data center initiatives made possible by enhanced use leasing, are a direct investment in Army priorities."

The US is narrowly ahead of China in AI development BUT HDC buildout is key

Lang 26 [Nikolaus, Sylvain Duranton, Vladimir Lukic, Matt Langionerigo Ortiz Mena, Jona Lampert, and David Zuluaga Martínez (all researchers @ BCG), 06-30-2026, "The Great Divide: How the US and China Are Splitting the AI World", BCG, <https://www.bcg.com/publications/2026/us-and-china-ai-strategy-causing-global-ai-divide>, accessed 07-12-2026] wrong //prav

The AI landscape today is dominated by the US and China as both countries continue to put the weight of their private and public sectors behind the development of this technology.¹ But despite the superpowers' shared recognition of the importance of AI as a source of national power, their AI strategies have diverged. Each country is developing an AI technology stack designed to reduce its dependence on the other—which also means their stacks are increasingly incompatible. CEOs and policymakers across the globe may have to choose sides sooner than they think.¶ In 2024, we argued that the US was the clear leader in the burgeoning AI

race, with China quickly gaining ground. Our updated, **2026 analysis across the six key enablers of AI supply—capital, talent, intellectual property (IP), data, energy, and compute—reveals the US has maintained its lead, fueled by its strength in talent and capital deployment.** (See “Methodology.”) But **China has continued to close the gap** on compute power and has **systematically translated** its IP strengths into a **consistent “fast follower” approach to frontier model development**, with clear focus on rapid adoption of AI across the real economy. (See Exhibit 1.) ¶ Methodology ¶ The US and China Are the AI Superpowers ¶ Weekly Insights Subscription ¶ Stay ahead with BCG insights on international business ¶ Enter Email ¶ Subscribe ¶ As the US and China pursue AI sovereignty, the effects of their bifurcating technology stacks will be felt across geographies and sectors—defining the options leaders have available to engage with AI. In our 2024 study, we found that countries and companies could still mix and match across both superpower stacks. Since then, the two ecosystems have diverged to the point where neutrality may soon become harder to sustain, which changes the calculus. For most countries, the question is how best to develop AI resilience by ensuring robust access. AI is no longer something countries compete for, but something they compete with. For senior executives, the choice of which AI stack to use increasingly determines where a company can operate and how exposed it is to geopolitical volatility. ¶ The US and China: The Great Bifurcation ¶ In late 2024 we argued that China had effectively reached parity with the US on AI model performance. Then, in January 2025, the US announced “Stargate,” a \$500 billion AI infrastructure joint venture led by OpenAI, Oracle, SoftBank, and MGX, while China released its highly capable and low-cost DeepSeek R1 model. ¶ These events were early indications of the increasingly distinct AI ecosystems that the US and China are committed to developing and expanding. China is increasingly focused on foundational research (publications, patents), domestic chip development, and real-economy adoption (deployment across industry and public services). By contrast, the US dominates the model and infrastructure layers, pouring capital into frontier models and the compute power required to serve them. ¶ The US: Building Data Centers and Top-Performing Models with Big Capital ¶ The **US strategy can best be described as winning through scale.** In practice, this has meant an **acceleration in capital deployment** over the past 18 months, with the **center of gravity shifting** from frontier model development **toward the data center infrastructure** needed to meet **growing inference** demand from enterprise uptake and the rise of AI agents. ¶ Even with this shift, **significant capital continues to flow into the development of frontier models.** Since 2023, US-based startups have raised around \$380 billion in AI-related venture capital, and incumbent tech giants invested more than \$300 billion in R&D in 2024 alone. A sizeable portion of those funds is invested in AI and adjacent innovations. The depth of these capital pools exceeds every other country’s and is reflected in the valuations of frontier model developers. (See Exhibit 2.) ¶ The US Leads in Capital Deployment for AI ¶ What’s new is the sheer scale of spending by US tech giants on the infrastructure to serve those models: CAPEX by the top technology companies surpassed \$400 billion in 2025—compared with \$63 billion in China—and is estimated to exceed \$800 billion in 2026. (See Exhibit 3.) That spending is expanding what is already the world’s largest data center base with 50+ GW of capacity at the end of 2025, compared to 31 GW in China and 12 GW in the EU. ¶ This extraordinary level of investment is responding to skyrocketing demand. For example, Google’s monthly inference volume grew about 50-fold year-on-year, and the three largest US cloud providers now hold more than \$1.4 trillion in contracted future revenue, more than double a year ago and in significant measure driven by AI. ¶ The US Speeds Ahead on Tech CAPEX ¶ The seemingly unstoppable rise in both CAPEX and VC investment in AI reflects the tightly integrated network of cross-investments and commercial commitments that has come to characterize the US tech ecosystem. Since 2024, **deals worth more than \$1.5 trillion have been announced between US AI labs, chip designers, hyperscalers, and capital providers:** equity stakes, **multi-year compute contracts,** and chip purchase agreements that link the major players to each other on multiple sides at once. (See Exhibit 4.) These investments are part of a concerted effort by big tech companies to ensure that suppliers across the AI value chain have the necessary capital to keep up with skyrocketing demand. While these integrated relationships and capital commitments are based on real, rapidly growing revenues, they create systemic risk in the US tech ecosystem: a default, write-down, or supply disruption at any major player could ripple quickly across the rest. ¶ The Self-Reinforcing US AI Ecosystem ¶ The inflow of capital to AI has been aided by the US government’s emphasis on promoting bundled, exportable “AI stacks” of chips, models, cloud, and software. While maintaining targeted restrictions, **US policy is more focused on outpacing through speed, scale, and ecosystem reach.** ¶ The US’s light-touch regulatory strategy, however, has been tested by the arrival of increasingly powerful models with national security implications. In April 2026, Anthropic announced it was delaying the public release of its Claude Mythos model to allow the US government and select companies to prepare for the model’s cyber capabilities. A version of Anthropic’s model with built-in safeguards was released to the public in June 2026; days later, the US government issued an export control directive for Anthropic to suspend all access to the model by any foreign national, whether inside or outside the US (with the net effect that the model had to be suspended for all customers), indicating that national security concerns may in some instances take precedence over the global diffusion of the most capable AI models. ¶ China: Prioritizing Adoption of Cost-Optimized AI ¶ China is pursuing a visibly different trajectory. While its investments remain sizeable, it is not mobilizing a comparable share of its economy to match the US at the frontier of model development or data center expansion. Instead, its strategy has focused on accelerating domestic adoption of AI while decreasing its reliance on foreign chips. ¶ This shift is premised on China’s ability to remain competitive at the level of foundation models despite its limited access to the highest-end computing power. Indeed, since the release of DeepSeek R1 in January 2025, top-performing Chinese models have kept up with US leaders. (See Exhibit 5.) Importantly, this isn’t just about DeepSeek; China has consolidated a proper ecosystem of AI startups and big tech incumbents developing models—including Alibaba, ByteDance, Moonshot, Z.AI, MiniMax, and Xiaomi. The open-weight nature of most Chinese models has helped consolidate this ecosystem, as labs build directly on each other’s weights and architectures. ¶ China Builds Powerful, Cost-Optimized Models ¶

China has stayed close to the US on model quality benchmarks while making its models multiple times cheaper than US counterparts, driven by open weights for most leading models and architectural efficiencies from novel model engineering. Moonshot's Kimi K2.6, China's top model in May 2026, runs at \$1.71 per million tokens, versus \$11.25 for GPT-5.5, the US's top model. These frontier advancements are underpinned by a deep and improving research base: China now accounts for roughly one third of the world's top 10% most-cited AI publications, ahead of the US, and is the clear leader in the number of AI patents, a reflection of its focus on economic applications of AI. (See Exhibit 6.)¶

Ferren 26 [Matthew Ferren; previously served in the Department of Defense where he developed strategies, policies, and plans for employing military cyberspace operations to advance U.S. national security interests and counter major cyber threats; *Council on Foreign Relations*; "The U.S. Is Losing the AI Credibility War—to Itself"; <https://www.cfr.org/articles/the-u-s-is-losing-the-ai-credibility-war-to-itself>; 6-22-2026] cruz

On June 11, Anthropic apologized after it emerged that its newest AI model, Fable 5, had been silently limiting responses to users suspected of attempting to replicate its technology. The model had also been criticized for refusing to respond to any cyber-related queries, redirecting users to less capable models instead. Two days later, President Donald Trump's administration barred foreign nationals from accessing Anthropic's two newest frontier models, Fable 5 and Mythos 5, citing national security concerns. Unable to screen users by nationality, Anthropic announced it had disabled both models worldwide.¶ The episode underscored how much both administration officials and frontier model developers now recognize the serious cybersecurity risks posed by advanced AI. For Anthropic's founders, AI safety and security have long been central to their mission. For the White House, Mythos's emergence earlier this year produced a remarkable about-face, forcing the administration to shift from an aggressive deregulatory agenda to one that is more risk-conscious.¶ Although this shared focus on AI and cybersecurity is a positive development, over-indexing on risk and failing to align on a clear way forward could cause the United States to miss out on a generational opportunity to improve national cyber defenses.¶ Using AI for defensive advantage.¶ The administration's post-Mythos strategy centered on a cyber window of opportunity. The White House believed that American companies and agencies could leverage advanced AI tools to fix as many software vulnerabilities as possible before adversaries acquired and weaponized similar capabilities. Frontier labs operationalized this approach by providing—and sometimes funding—limited access to models with advanced cybersecurity capabilities.¶ But two recent developments reportedly drove the White House to abruptly reverse course: first, that restricted models had been accessed by unauthorized parties, and second, that guardrails in public models could be bypassed in limited cases. The restrictions on Anthropic's models are now causing widespread uncertainty in the cybersecurity market and risk wasting a fleeting opportunity to harden U.S. networks.¶ AI tools promise to shift the dynamics of cyber defense and exploitation in favor of defenders. Realizing this outcome requires putting these capabilities in the hands of as many defenders as possible, even if malicious actors also gain access along the way. Transparency has long been a foundational principle of information security. Open-source software and bug-bounty programs provide shared resilience, even when they surface vulnerabilities along the way.¶ Adversaries will eventually obtain their own advanced AI capabilities. That may come from jailbreaking publicly available frontier models, gaining illicit access to controlled U.S. models, or developing their own. Estimates put China's leading models at roughly three to eight months behind the most advanced U.S. models. Even if U.S. labs preserve their relative advantage, Chinese offensive cyber capabilities will continue to mature.¶ When China acquires its own Mythos-like capability, the United States will need to be ready. That means maximizing this window of opportunity to expand access to defensive tools and increase the speed and scale of software patching.¶ Controlled access programs like Project Glasswing are a strong start, but frontier labs aspiring to b9

Constant development is key---the AFF decks it, wrecking deterrence and losing the AI race.

Deptul 26 [David A. (dean of the Mitchell Institute for Aerospace Studies), 05-28-2026, "Blocking the construction of data centers is a national security risk", The Washington Post, <https://www.washingtonpost.com/opinions/2026/05/28/data-centers-are-critical-ai-military-national-security/>, accessible @ <https://archive.is/dtzuX>, accessed 07-12-2026] wrong

Blocking the construction of data centers is a national security risk” Debates over data center

construction have generally targeted the issue as a local zoning nuisance or a threat to the environment and affordable electricity. But this **overlooks a crucial fact**: The **rapid construction of data centers** and advanced energy infrastructure required to power them **are foundational to American military power** and must be a national security priority. ¶ **Winning the wars of the future will require long-range precision weapons, advanced combat aircraft, resilient space systems, sophisticated missile defenses, and autonomous systems at sea, land and air — plus the data infrastructure to connect, train and sustain them all.** ¶ That last requirement is too often ignored. ¶ The United States cannot prevail in the next era of military competition with assumptions about data infrastructure from the 20th century. Data is no longer merely a tool of commerce. It is a strategic asset. Nearly **every function in the military depends on the ability to store, move, process, secure and exploit vast quantities of data at speed and scale.** ¶ A **nation** that cannot **store and process data will not deter** — nor, if necessary, defeat — a peer **adversary**. ¶ Many communities are understandably worried that data centers are increasing electricity costs, water use, carbon emissions and noise. Opposition to their construction has become a bipartisan issue. But **if the nation is to dominate the development of artificial intelligence, autonomous operations or the future of intelligence gathering, it needs the physical infrastructure.** ¶ The U.S. is the **global leader in data center capacity**. But China has made data infrastructure a strategic priority and is **building capacity at astonishing speed**. By 2030, it could **easily double its current capacity**, even while still trailing the U.S. ¶ **China is expected to double its data center capacity by 2030**. ¶ China's cumulative data center capacity ¶ Traditional AI and high-performance computing ¶ 2020 ¶ 2021 ¶ 2022 ¶ 2023 ¶ 2024 ¶ 2025 ¶ 2026 ¶ 2027 ¶ 2028 ¶ 2029 ¶ 2030 ¶ 0 ¶ 5 ¶ 10 ¶ 15 ¶ 20 ¶ 25 ¶ 30 ¶ 21.2 ¶ 30.9 ¶ 10.5 ¶ 28.9 ¶ Source: Rystad Energy ¶ Those who doubt China's ability to build quickly need only look at recent history. The country has laid thousands of miles of high-speed rail since 2008, its manufacturing sector is nearly double the size of America's, and China dominates the production of consumer electronics, electric vehicles, advanced nuclear reactors, solar panels and many other high-tech products. Militarily, China already has the world's largest navy and may have a fighter air force as large as the U.S. by 2028. When China's State Council says it plans to “build an infrastructure that supports large-scale data circulation and interconnectivity,” the U.S. should take note. ¶ **A shortfall in data storage and computing capacity could be catastrophic. Success in future warfare will depend on whether a belligerent has the capability to sense, decide and act faster than an adversary.** That **requires enormous quantities of intelligence and reconnaissance, cyber, logistics, targeting and operational data. It requires computing capacity to train AI on all of it.** ¶ Ukraine has already demonstrated that the **wars of the future will in part be fought by autonomous and remotely operated systems.** The **side that can connect data to decisions faster and more reliably than its adversary will be the one that wins on the battlefield.** ¶ In a future operation, **data processing will affect everything from missile defenses and autonomous aircraft to deployment logistics and intelligence storage and analysis.** In that environment, **data infrastructure becomes critical to deterrence.** Recent **attacks by Iran on cloud and data infrastructure in the Middle East have only underscored how data centers are nodes of national power.** ¶ **Americans** do not want higher utility bills or strain on the environment, nor do they want poorly planned industrial development imposed on their communities without their consultation. But **they also wouldn't want China to win the AI race or to see U.S. forces outpaced in a fight over the Taiwan Strait.** ¶ **If the U.S. is serious about preserving military superiority over China and other adversaries, then data must be treated as the critical asset** it is while explaining to the public why it is needed. **To build this infrastructure effectively, the approval processes required to build must be streamlined**. It also needs power, which means building more energy capacity and strengthening the electrical grid. And to ensure its security and resilience it must be decentralized, which means data centers must be dispersed across the country — they can't all be in Northern Virginia, as so many are. ¶ The nation with the best data infrastructure will possess a decisive advantage in the next era of warfare. The U.S. cannot afford to lose that position.

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And lest we be overconfident in Silicon Valley, let's remember the activists. A small group of activists shut down supersonic technology, and a small group of activists shut down all nuclear clean energy in this country. Okay. We have 100 fission reactors being built in China. We have one in the United States. It's a disaster that happened. And so we can't take for granted that the kooks who are calling for data center moratoriums. Right. Which just think about this for a second. All of our GDP growth is coming from the fact that we are building data centers and driving AI and driving productivity improvements in the economy. A data center moratorium would thrust us straight into a recession and high unemployment. Secondly, it would cede the entire global game to China. Like overnight we would lose to China in the global air race, which is not just about AI, but it's about economic security, it's about jobs and it's about national security. So it literally is insane that we would do this. I can't even believe there are people talking about it. However, why are they talking about it? Because people are concerned. Local communities are concerned. I just got back from celebrating my mother's 90th in rural Indiana, you know, over the Memorial Day weekend. And what I'll tell you.

The AFF prompts Chinese invasion.

Wuthnow '23 [Joel; 3-24-2023; Senior Research Fellow in the Institute for National Strategic Studies at the National Defense University; Foreign Affairs; "How to Out-Deter China,"

<https://www.foreignaffairs.com/united-states/deter-china-taiwan>] TM + Leon

But in the years since, China has worked to make a U.S. intervention less likely through an approach it calls “strategic deterrence,” which relies on, among other things, using nuclear signals to dissuade a potential adversary from entering the fray. China’s deterrence efforts are intensifying even as the Biden administration moves ahead with its own plans for the “integrated deterrence” of Chinese aggression, which involves threatening military and economic penalties in concert with a coalition of allies to convince China of the tremendous costs of war. These two competing models of deterrence are at odds with each other in ways that could destabilize the Taiwan Strait and the region at large. China, spurred by its perception of U.S. decline, emboldened by its rapidly expanding nuclear arsenal, and inspired by Russian President Vladimir Putin’s apparent success in using nuclear threats to limit U.S. support for Ukraine, could become overly confident and spark a conflict in the belief that Washington will stay out of the way. Washington must avoid this kind of escalatory spiral by undermining Chinese optimism in its own capabilities; in other words, by out-detering China. This requires delivering an unequivocal message to Beijing that any conflict between the two nuclear-armed powers could quickly become calamitous, far outweigh[ing] the potential benefits of an armed reunification with Taiwan If deterrence fails—if China grows more convinced of its military superiority and underestimates the U.S. commitment to the island—both countries could end up embroiled in a war between great powers armed with nuclear weapons.

Growth of AI development is key

Krawec '26 [Christina Krawec is the founder of Earthnote LLC and a former analyst at the International Atomic Energy Agency, U.S.

Department of Defense, and Google. She holds a master's degree in Non-proliferation and International Security from King's College London, 5-12-2026, "Tracking Hyperscale AI Data Center Growth with Satellite Imagery", Federation of American Scientists, <https://fas.org/publication/tracking-hyperscale/>, Accessed 7-5-26] // shang

The advancement of AI is inextricably linked to its physical foundations. Hyperscale data centers power the world’s most advanced AI models, and the demand for these facilities continues to accelerate. Governments and frontier AI developers alike are investing tens to hundreds of billions of dollars into hyperscale data center buildouts, sparking debates on the economic, environmental, and geostrategic impacts of such efforts. On the international stage, AI is increasingly considered a source of economic growth and an asset to military capabilities resulting in global competition, particularly between the U.S. and China. In addition, the successful operation of data center facilities requires high quantities of energy, affecting local communities and the environment. Therefore, from grassroots community impacts to global geopolitical dynamics, understanding developing data center capacities is emerging as a critical analytical challenge.¶ The speed at which AI models are advancing necessitates the rapid growth of its supporting data center infrastructure. In tracking data center construction and operation, what can remote sensing—particularly, the use of electro-optical (EO) satellite imagery—meaningfully illustrate about data center development, and what can policymakers learn from this data moving forward? While multiple research organizations have utilized satellite imagery to track construction and ongoing activity at hyperscale data center facilities, no baseline assessment of its benefits and limitations is available in the public sphere. This research addresses that gap with two case studies: Khazna’s

Ajman data center facility in the United Arab Emirates (UAE) and xAI's Colossus data center site in Memphis, Tennessee. These examples reveal inconsistencies between commercial announcements and on-the-ground activity visible in satellite imagery, showing that official construction timelines, infrastructure permits, and company publications cannot be relied upon exclusively for insights into data center development. Satellite imagery provides a layer of independent verification that can be integrated to overall assessments of data centers, and this study illuminates where this imagery adds insight to inform future AI policy efforts. In particular, as discussions of international AI governance evolve, satellite imagery can be integrated as one verification method in a wider toolkit supporting the monitoring and implementation of future AI-focused agreements.

Large gaps in AI dev cause nuclear war

Pimpale 26 [Govind Pimpale; research fellow at the Foundation for American Innovation, where he focuses on AI policy. Before joining FAI, he worked at a startup developing reinforcement learning environments, and prior to that, as an AI evaluations researcher. He holds a BS in Computer Science and Engineering from UCLA; AI Frontiers; "An AI Capabilities Gap Can Endanger Nuclear Deterrence"; <https://ai-frontiers.org/articles/an-ai-capabilities-gap-can-endanger-nuclear-deterrence>; 6-25-26] //prav

Since the late 1950s, Mutual Assured Destruction (MAD) has served as a limiting factor on great-power conflict. The doctrine holds that, if two opposing nations have nuclear weapons that can survive one another's initial strike, then the near-certainty of devastating retaliation will deter each side from launching a large-scale nuclear attack. Despite this logic, military planners have long considered the possibility of a counterforce nuclear attack, where a superpower uses nuclear weapons to cripple the nuclear capabilities of its enemy. If such an attack were executed preemptively, as a so-called "first strike," it could both start and end a great-power conflict in a matter of hours: without retaliatory capacity, the defender would be at the mercy of the aggressor's remaining nuclear weapons and forced to surrender.¶ The reason this does not happen is that a truly successful counterforce strike is nearly impossible to pull off: a would-be attacker doesn't know the locations of all opposing missile launchers and submarines, nor does it have missiles with sufficient precision and speed to destroy opposing missile silos before they could launch retaliatory nuclear warheads. Thus, if provoked by a first strike, the opposing side would likely be able to launch a large-scale nuclear response, and the attacker, unable to counter all enemy missiles, could face devastating losses. The result of a preemptive counterforce strike, in other words, would be mutual assured destruction.¶ AI could change this dynamic. By the mid-2030s, AI-assisted research and development could reduce the cost required to develop and manufacture military hardware by an order of magnitude. Such lower costs could enable a nation with an AI lead to quickly and cheaply build military infrastructure projects, making nuclear counterforce strikes a realistic possibility. Nations with weaker AI capabilities would struggle to quickly build the countermeasures needed to retain a credible nuclear deterrent.¶ << TEXT CONDENSED; NONE OMITTED >>¶ This article will primarily be a technical assessment of how AI could undermine nuclear deterrence, although I'll lightly touch on a few political aspects. For a longer and more in-depth assessment of the factors discussed here, consider taking a look at an earlier piece I wrote.¶ Why MAD Has Worked So Far¶ In any preemptive counterforce strike scenario, the attacker is at a large disadvantage compared with the defender, because they must destroy all of the defender's nuclear weapons. To execute a counterforce strike, the attacker must achieve all the following requirements simultaneously:¶ Suppress launch on warning. Many nuclear-armed nations have a policy of launch on warning—launching a retaliatory strike based on sensor data, before a nuclear weapon has been confirmed to land on their territory. The attacker must either use fast-arriving weapons or otherwise disable launch on warning.¶ Locate and destroy nuclear submarines. Nuclear submarines, while extremely stealthy, are vulnerable once detected. Each nuclear submarine can carry and launch hundreds of warheads. The attacker must accurately track their locations in real time and destroy them all within minutes.¶ Locate and destroy mobile launchers. China and Russia (although notably not the US, the UK, or France) each field a set of mobile missile launchers that can be dispersed during times of high alert. These mobile launchers are extremely difficult to locate for long enough to successfully strike. However, if accurate and up-to-date position data can be provided, they are easy to destroy.¶ Destroy all silos. Hardened silos have known positions, but they require almost a direct hit—ideally with a nuclear weapon—to ensure their destruction. They stand out among nuclear launch platforms, as they have the fastest reaction time and the hardest-to-disrupt communications.¶ Defend against surviving missiles. If the attacking nation has a functioning missile defense system, it may not need to destroy all of an enemy's nuclear weapons, as its missile defense can handle some leftovers. The stronger a nation's missile defense, the less thorough its first strike has to be.¶ Overcome the nuclear taboo. One theory for why nuclear weapons are rarely used is the nuclear taboo. The leaders of both the US and the USSR recognized the gravity of nuclear weapons usage, and had serious humanitarian reservations about starting a nuclear conflict. Even if a counterforce strike were perfect, radioactive nuclear fallout was expected to kill tens of millions in the target country.¶ The six requirements above have thus far preserved deterrence not because they are physically impossible to meet but because meeting them at the scale required has always been prohibitively expensive.¶ A useful paradigm to consider in nuclear conflict is the cost-exchange ratio. When one side fields an additional weapon, how much must the other side spend to neutralize it? The concept comes from the Cold War debate over ballistic missile defense. If an attacking nation can

build another nuclear missile for \$1 million, and the interceptor needed to stop it costs the defending nation \$10 million, then missile defense is a losing game: each \$1 spent on offense forces a \$10 expenditure on defense. This particular cost-exchange ratio explains why a nation launching a counterforce strike would struggle to defend against surviving missiles.¶ << PARAGRAPH BREAK¶ S CONTINUE >>¶ Why

AI Could Dramatically Cut Military Infrastructure Costs¶ AI-assisted military R&D could invert the cost-exchange ratio. If **a nation with an**

AI lead can automate most of the **design,** systems **integration, and production** labor that currently makes military

infrastructure expensive, that nation's costs could be significantly reduced while its enemies' costs stayed the same (since they

don't have the AI advantage). If interceptors cost only \$100,000 and missiles still cost \$1 million, then it suddenly becomes rational to build more interceptors.¶ The likelihood of this prediction's coming true rests on answers to two questions: (1) whether AI can automate most of the intellectual labor, and (2) whether doing so really cuts costs by a margin significant enough to change militaries' economic calculus. Notably, this forecast does not require **progress in robotics, which could further reduce costs.** I'll focus on aerospace

manufacturing, which comprises most of what the attacker needs to build: new missiles, surveillance constellations, and missile defense systems.¶ Can AI automate most intellectual labor? Aerospace projects of the type I'm describing here are highly

interdisciplinary, and even simple systems demand expertise in electrical, computer, and mechanical engineering. More-advanced projects may require fundamental research in applied physics or math. While AI assistance for software development is a relatively mature use case, AI abilities in the other fields are much more nascent.¶ Yet there is good reason to believe that most intellectual fields will follow the same trends. AI abilities have increased at roughly the same rate across various domains, as shown in the chart below.¶ << CHART OMITTED >>¶ Additionally, we're already beginning to see AI labs show interest in automating science and engineering fields: Anthropic is training on electrical engineering, and the company showcased 3D modeling performance in its Claude Fable 5 launch post. OpenAI has made discoveries in mathematics and physics. On the biology front, Google DeepMind's AlphaFold has solved protein folding, enabling advances in drug discovery.¶ Extrapolating from such innovations, one engineer could do the work of many: outsourcing most of it to agents and handling only what agents can't yet do, like running experiments or meeting stakeholders in person.¶ How much can automated intellectual labor lower military R&D costs? Although aerospace manufacturing seems like a labor-heavy job, it's a remarkably white-collar profession, with high exposure to AI automation. In a 2021 article, Princeton University computer scientist Edward Felten and co-authors calculated that aerospace manufacturing has an AI Industry Exposure Score of 0.519, in the 70th percentile of all industries (around the same level as real estate). If AI automates the intellectual labor of aerospace manufacturing, this would substantially lower the cost of the final product.¶ To make things more concrete, consider SEC filings showing the financials of two public launch services companies, Rocket Lab and Firefly Aerospace.¶ << TABLE OMITTED >>¶ R&D (which accounts for 28%–48% of spend at the two companies) is mostly engineering compensation and analysis, with the remainder going to propellant, test facilities, and hardware. It's mostly cognitive labor, and therefore a potential target of future AI automation. SG&A expenses (21%–22% of spend) describe the cost of finance, legal, contracts, compliance, and business development. These disciplines rely almost completely on cognitive labor. Cost of revenues (31%–51%) is the most mixed category: it blends physical labor (technicians doing welding, assembly, and launch operations) with white-collar labor (manufacturing, quality, and test engineering), and purchased materials and components.¶ The costs of aerospace components are themselves quite compressible. For example, star trackers (devices used by satellites to measure their own orientation) can cost over \$100,000. These devices' physical components—a camera sensor, a housing, and some processing hardware—may cost as little as a few thousand dollars altogether. The more significant cost comes from qualification testing and R&D (because of low manufacturing volume, such costs can't effectively be amortized). So, by reducing the cost of R&D, automating intellectual labor can drive down component costs.¶ I expect the cost savings to be even greater for defense aerospace products, which usually cost more than comparable commercial products. This higher cost partly reflects the increased documentation and security practices required for highly regulated uses.

Such regulatory compliance practices rest almost entirely on cognitive labor that can be automated. Another driver of these product's higher costs is costlier labor: defense contractors must have security clearances, limiting the worker pool and raising wages. An AI, on the other hand, must be cleared only once; it can then be scaled indefinitely.¶ What Could We Build?¶ Recall the six requirements that have kept a counterforce strike out of reach. Each one has held not because it was physically impossible, but because meeting it at scale was prohibitively expensive. Once cheap intellectual labor collapses those costs, an attacker with an AI lead could pursue several military megaprojects at once, each one solving a different requirement needed to launch a successful counterforce strike. This section will focus on the three requirements I expect to become significantly easier to meet.¶ Locating and destroying nuclear submarines. Research has shown that, in certain conditions, moving submarines leave a surface wake detectable by Synthetic Aperture Radar (SAR), a space-based radar system. A large constellation of SAR satellites would permit near-continuous surveillance of the entire world, regardless of time of day or local weather conditions. The US is already pursuing a SAR satellite constellation, and will launch its first satellite in 2028. The US could supplement this effort with a space-based constellation of Light Detection and Ranging (LiDAR) satellites. Each satellite would measure depth via pulses of intense light, detecting even stationary submarines (although that would require clear weather). Certain frequencies of LiDAR can detect submarines within 200 meters of the surface, within typical nuclear submarine operating depths. A dense constellation of LiDAR and SAR satellites could sweep the oceans, and expose submarines at scale.¶ However, while both satellite methods can reveal submarine positions temporarily, the

only tool to keep track of them consistently would be underwater drones (often called Unmanned Underwater Vehicles, or UUVs). UUVs are currently limited by the difficulty of autonomous operation, but AI R&D would likely significantly

improve this. The US has been funding research in this direction.¶ Defending against surviving missiles. A system with tens of thousands of interceptors prelaunched in space (similar to the 1980s Brilliant Pebbles concept, originally abandoned due to cost issues) could counter any missiles that are missed by the first strike. The US is already pursuing this, too, with its Golden Dome system.¶ Overcoming the nuclear taboo. The missiles of the Cold War had poor precision, so warheads with an explosive yield of hundreds of kilotons were common (designed to

guarantee a silo kill). But, as mentioned, such heavy warheads would result in enough fallout to guarantee millions of deaths. In 2017, Keir A. Leiber of Georgetown and Darryl G. Press of Dartmouth found that modern missiles have much higher precision than those from the Cold War, and future improvements could reduce the average targeting error to mere meters. With such high precision, very low-yield weapons could be used, with little to no fallout. A counterforce strike could be accomplished with an estimated death toll of around tens of thousands rather than millions, well within the range of wars nations are willing to start.¶ Other aspects of the retaliator's deterrent are vulnerable to AI R&D too. The retaliators' **mobile missile launchers** can be located with the exact same SAR constellation that we discussed for submarine detection, leaving them vulnerable to a first strike. **Launch on warning systems are also vulnerable.** While it's unlikely they could be hacked outright, Anthropic's **Mythos demonstrated AI driven vulnerability discovery** that **could be used to confuse, delay, or reduce confidence.** Additionally, the defender's early warning response time can be shortened significantly with depressed trajectory submarine-launched missiles, which could cover 2000 km in only 10 minutes. The **combination of high-speed and precise missiles also works to efficiently counter fixed silos.**¶ The Defender's Options¶ If a lagging nation realizes that a rival is on course to achieve nuclear primacy—the ability to execute a counterforce strike without retaliation—it still has a few options.¶ The first, and most straightforward option is to expand the arsenal. It can increase the number of silos, build more nuclear submarines and mobile missile launchers, and raise its level of alert. This strategy would work in the short term, since the attacker would be forced to scale up its own forces until it was certain it could neutralize all of the new forces. The problem is the cost-exchange ratio. If the attacker has an AI advantage, the retaliator could end up paying a larger price for each new silo than the attacker pays to build the missiles or interceptors that could defeat it.¶ The second option is to target the root cause: the AI gap between the attacker and retaliator. Potential avenues in this direction can range from relatively diplomatic to **highly escalatory.** Options include disrupting the attacker's supply chain, launching data poisoning attacks, or sabotaging its AI training runs. However, the most extreme actions—**direct kinetic attacks** on datacenters or researchers—**would be likely to start wars.** But even these interventions will only be effective if applied early. Once the lead is large enough, sufficiently smart AIs will already be trained. Thus, an AI-lagging defender must be alert enough to act before the AI capability gap becomes overwhelming.¶ Finally, the attacker and retaliator could negotiate a treaty. A bilateral arms-control regime could in principle cap satellite constellations, ballistic missile interceptors, or AI compute used for military R&D. There is precedent here, especially in the nuclear domain. However, the main challenge will be aligning incentives. **The leading nation has no incentive to join a treaty where only the lagging party stands to gain.** Traditional arms-control treaties have only worked where there were symmetric costs on both sides

The AFF also prompts Chinese invasion of Taiwan

Wuthnow '23 [Joel; 3-24-2023; Senior Research Fellow in the Institute for National Strategic Studies at the National Defense University; Foreign Affairs; "How to Out-Deter China," <https://www.foreignaffairs.com/united-states/deter-china-taiwan>] TM + Leon

But in the years since, China has worked to make a U.S. intervention less likely through an approach it calls “strategic deterrence,” which relies on, among other things, using nuclear signals to dissuade a potential adversary from entering the fray. China's deterrence efforts are intensifying even as the Biden administration moves ahead with its own plans for the “integrated deterrence” of Chinese aggression, which involves threatening military and economic penalties in concert with a coalition of allies to convince China of the tremendous costs of war. These two competing models of deterrence are at odds with each other in ways that could destabilize the Taiwan Strait and the region at large. **China, spurred by its perception of U.S. decline, emboldened by its rapidly expanding nuclear arsenal, and inspired by Russian President Vladimir Putin's apparent success in using nuclear threats to limit U.S. support for Ukraine, could become overly confident and spark a conflict in the belief that Washington will stay out of the way.** **Washington must avoid this kind of escalatory spiral by undermining Chinese optimism in its own capabilities;** in other words, by out-detering China. **This requires delivering an unequivocal message to Beijing that any conflict between the two nuclear-armed powers could quickly become calamitous, far outweigh[s]ing the potential benefits of an armed reunification with Taiwan.** If deterrence fails—if China grows more convinced of its military superiority and underestimates the U.S. commitment to the island—**both countries could end up embroiled in a war between great powers armed with nuclear weapons.**

China would invade Taiwan to maintain its lead.

Tucker 25 --- (Patrick Tucker [Science and Technology Editor], 9-18-2025, "How the rapid pace of AI connects to China's threats toward Taiwan", Defense One, <https://www.defenseone.com/technology/2025/09/what-rapid-pace-ai-means-chinas-threats-toward-taiwan/408220/doa7-18-2026>) //nl

Ten years ago, if you said artificial intelligence might surpass human-level intelligence by the year 2029, no one in Washington would have taken you seriously. A Wednesday hearing showed how dramatically that has changed. During the hearing, dubbed "Shaping Tomorrow: The Future of Artificial Intelligence," the House Oversight Committee and its witnesses grappled with several trends: **the rapid advance of AI,**

Beijing's determination to dominate the field, and China's increasingly threatening posture toward

Taiwan. "Current trend lines suggest" that Ray Kurzweil—inventor, author, and current Google executive—"was dead on" when he predicted human-level AI by 2029, Samuel Hammond, chief economist at the Foundation for American Innovation, said at the hearing. Kurzweil's projection for the arrival of artificial general intelligence, or AGI, has been a staple of Silicon Valley cocktail parties for more than two decades. The prediction is not universally endorsed by technologists; dissenters include Meta's Yann LeCun and Google Brain founder Andrew Ng. But fans of AGI-by-2029 include Elon Musk and Google DeepMind creator Demis Hassabis—and increasingly, state-run Chinese media. Technologists have typically discussed it with a sort of gleeful fascination. But the hearing was anything but optimistic, especially when the subject **focused on the AI race** with China. If **China's government, which has already dedicated the country to ambitious AI goals, expects AGI within four years, the country's increasing harassment of Taiwan** takes on an even more menacing dimension. At the hearing, Hammond said **a Chinese takeover of Taiwan would spark a "global depression," because Beijing would choke off the flow of advanced computer chips from the island.** That echoes warnings from former Biden Commerce Secretary Gina Raimondo, the conservative Heritage Foundation, and others. **Chips made in Taiwan, particularly by world leader TSMC, are key to the world's manufacturing and economic activity—and will only grow more so in an era dominated by advanced artificial intelligence.**

Invasion would lead to nuclear war.

Brands-24 [Hal Brands, senior fellow @ the American Enterprise Institute and the Henry A. Kissinger Distinguished Professor of Global Affairs @ the Johns Hopkins School of Advanced International Studies, 8-28-2024, If South Korea Goes Nuclear, So Will the World, American Enterprise Institute, <https://www.aei.org/op-eds/if-south-korea-goes-nuclear-so-will-the-world/>] BZ

A second factor is aggression by nuclear-armed predators against non-nuclear prey. Russia has brutally assailed Ukraine, which gave up its nuclear weapons in the 1990s. Moscow then used its arsenal to deter the US from intervening directly. That Ukraine has held its own in this war has limited the global fallout. But a world in which countries with the bomb repeatedly brutalize countries without it will quickly become a far more nuclearized world. **If China was to invade Taiwan — and the US was unable or unwilling to stop it —**

proliferation pressures could increase dramatically. The final factor, and what most concerns South Korean officials — a potential American withdrawal — is what would most devastate the non-proliferation regime. As long as US alliances are strong and credible, US allies have better, cheaper options than nuclear self-help. Even if Iran goes nuclear, for instance, stronger US security guarantees for Saudi Arabia — and support for the kingdom's civil nuclear program — can probably keep it from doing likewise. **But if the US pulls back, erstwhile allies from Eastern Europe to East Asia might feel that they face a choice between nuclear proliferation and national suicide** — which is why debates about acquiring those weapons have gotten louder in the age of Trump. If Trump wins in November, he probably won't actually quit US alliances: Then he would lose the ability to gripe about them. But he will continue to sow doubt about America's geopolitical commitments, and some of his once-and-perhaps-future advisers say proliferation by US allies wouldn't be so bad after all.

That locks in unsafe AI from authoritarian governments

Phillips-Robins 26 [Alasdair Phillips-Robins (J.D. from Yale, Fellow @ Carnegie Endowment), Teddy Tawil (AI Specialist and Visiting Researcher @ Carnegie Endowment), Sam Winter-Levy (PhD in Politics from Princeton, Senior Fellow @ Carnegie Endowment), 06-08-2026, "The Compute Coalition: How to Build the Future of AI in the Free World", Carnegie Endowment for International Peace, <https://carnegieendowment.org/research/2026/06/the-compute-coalition-how-to-build-the-future-of-ai-in-the-free-world>, DOA: 7-14-2026] |prav

Introduction¶ Across the world, one of the largest peacetime industrial mobilizations in history is under way. This year alone, America's biggest technology companies will spend some \$670 billion, or about 2 percent of U.S. GDP, building compute clusters. Worldwide, companies and governments will pour almost \$1 trillion into data centers with a single goal: building transformative AI.¶ **The United States currently dominates** the buildout: As of May 2025, almost **three-quarters of the world's advanced AI computing clusters were on American soil.** U.S. projects also move faster than those in most other countries. **But that lead is fragile. Domestic**

constraints—grid capacity, permitting rules, political opposition—are **tightening**. Abroad, China is **mobilizing to close the gap**, while Gulf states are touting energy and capital to attract developers.¶ Many traditional U.S. allies, meanwhile, risk being left behind. Across Europe, every major publicly reported AI data center combined appears to contain less computing power than the Amazon-Anthropic mega-cluster at New Carlisle, Indiana. Some U.S. allies, including Australia, Italy, and South Korea, have no major publicly known operational AI chip concentrations at all. And future plans, such as Australia and France’s gigawatt-scale projects, will at best match what industry leaders built years earlier.¶ That failure is **more than a question of economic returns**. The **countries that host the buildout will shape the future of AI: who controls it, what values it contains, how it is used**. If democracies are in the lead, they will have a shot at ensuring that transformative AI is safe, secure, and reflects liberal values. If authoritarian powers pull ahead, they will gain the tools to reshape the world order in their image, through repression at home and military dominance abroad. And because authoritarian regimes are more likely to suppress dissent and hide failures, an AI race they lead raises the risk of catastrophic accidents.¶ The free world still has the energy, capital, talent, and industrial might to win. This paper, built on a detailed new financial model of data center competitiveness, explains what governments can do. The model considers a wide range of factors, including construction, IT, and other capital costs; operating expenses; revenues; tax systems; and operational timelines. It helps explain what is driving billion-dollar company decisions, and how governments can change them.¶ The core finding: **time to power is what matters most**. In policy circles, the conventional wisdom often emphasizes energy costs and tax incentives. Our model suggests these factors are secondary. Countries that can get data centers online quickly produce dramatically better returns than those where projects languish in permitting and grid connection queues. We estimate that **a one-year delay in operation would cost an illustrative 100-megawatt U.S. data center more than \$500 million over its life cycle**, or more than 5 percent of its total value (see figure 1). Put differently, companies should be willing to pay at least double U.S. power prices to operate a data center just one year sooner.¶ The United States and the United Arab Emirates (UAE) top our country competitiveness rankings, in large part because major projects move fastest in those countries (see figure 2). By contrast, the country where timelines are longest, Germany, fares worst. But these rankings can shift quickly: with a one-year delay, the United States drops to fifth, behind the UAE, Finland, Canada, and India. If India sped things up by a year, it would jump ahead of Finland, Canada, and the United States, rising from fifth to second place. If war in the Middle East slows the UAE by a year and a half, it drops to fourth.¶ **Other factors matter, but none matches the significance of speed**. Even **doubling electricity prices would cost a U.S. facility less than a one-year delay**. Moderate tariffs on **GPU** servers would cost a U.S. facility about **one-third as much**. And removing typical state tax incentives comes in at about 60 percent of the cost of such a delay. Because IT equipment—priced similarly worldwide—accounts for the large majority of capital expenditures, national differences in non-IT factors, such as labor and land costs, have a small effect on overall project economics, and subsidies must be very large to move the needle significantly.¶ Our financial model focuses on what individual countries can do to compete. But no single democracy—not even the United States—can build the world’s AI infrastructure alone. A broader coalition of democracies, one that pools its geography, power generation, capital markets, and supply chain strengths, would be stronger, more resilient, and better positioned to ensure that transformative AI is developed and governed on democratic terms.¶ Democracies thus need to both compete at home and cooperate abroad. This paper makes recommendations on both fronts. On the domestic level, it identifies the reforms that would most improve countries’ competitiveness in attracting AI infrastructure. On the international level, it makes the case for a broader democratic coalition that can collectively outcompete authoritarian alternatives across the AI supply chain. The need is not for an overall acceleration in the AI buildout, but for a shift in relative progress to keep the center of gravity in the democratic world.¶ Domestically, the priority for countries that struggle to attract investment today should be removing the obstacles that slow projects down, including permitting backlogs, grid connection queues, transmission bottlenecks, and equipment shortages. Governments should not spend scarce public funds on a lucrative industry through energy subsidies, tax cuts, or other fiscal outlays. Rather, they should consider:¶ • Creating fast-track review processes for data center projects that meet certain criteria, such as the use of clean energy, investment in grid infrastructure, or tax commitments. As long as individual governments can actually shorten approval timelines, they will have significant flexibility to shape the bargains they strike with data center developers.¶ • Promoting grid flexibility and resilience by reforming interconnection queue processes, allowing operators to plan for load flexibility under the right circumstances, and encouraging the production of bottlenecked equipment.¶ • Supporting clean behind-the-meter power, including solar microgrids and wind farm offtake, as a bridge to grid connections.¶ Internationally, the democratic world needs to start building the partnerships that will let it collectively dominate the AI value chain from natural resources to last-mile deployment. Fruitful collaboration could include mutual fast tracks for allied infrastructure investment; common industrial policy in sections of the supply chain, such as critical minerals, where chokepoints lie outside the democratic world; joint safety and security research on frontier AI capabilities and safeguards; and common transparency and incident reporting standards. Together, democracies can set the pace in the development and security of powerful AI systems.¶ This paper proceeds as follows. Part I explains the economics of data centers: it lays out the findings from our model, shows that time to power is a key driver of companies’ decisions about where to site data centers, and examines what drives differences in competitiveness between countries. Part II explains the geopolitical stakes. Those siting decisions will help determine the future of AI and the balance of global power. The paper thus makes the case for a democratic compute coalition, arguing that the United States and its partners each have strong incentives to cooperate. Part III outlines the policy agenda: domestic reforms to reduce time to power and international coordination to secure the AI supply chain.¶ I. The Economics of Compute¶ **Companies are scouring the earth for land to build AI** chip clusters and energy to run them. Data center **economics show that the places getting chips online the fastest have a decisive advantage**.¶ The Model¶ In June 2025, two major AI infrastructure projects hit milestones. In Abilene, Texas, OpenAI’s Stargate data center took delivery of its first Nvidia chips. On the other side of the Atlantic, in Brussels, the European

Commission received seventy-six nonbinding “expressions of interest” for its AI Gigafactories initiative—an early step in its plan to support large computing clusters across the bloc.¶ By December, Stargate Abilene was running an estimated 100,000 Nvidia GB200 chips, or more than 200 MW of AI computing capacity. The commission, meanwhile, was still preparing to issue its formal call for proposals—a step that has since been delayed. More than a year after the program was announced, no construction has begun.1¶ OpenAI is not the only American firm forging ahead while European leaders issue paperwork. A single American company, Google, holds roughly one-quarter of global AI computing power. In 2025, the United States hosted around 75 percent of the world’s high performance AI compute. The European Union has less than 5 percent.¶ To understand how governments might improve their lot, we built a financial model to explain the major drivers of investment decisions. The model projects the life cycle value of a hypothetical 100 MW AI data center—large, but no longer frontier-training scale2—in ten countries, representing a range of players in the infrastructure race: Australia, Canada, Finland, France, Germany, India, South Korea, the United Arab Emirates, the United Kingdom, and the United States.¶ The model considers capital expenditures (IT equipment, construction, electrical systems, liquid cooling, on-site power generation), operating expenses (electricity, maintenance, staffing), tax treatment (corporate taxes, depreciation schedules, state and local incentives), time to operation, and projected revenues over a twelve-year life cycle, drawing on public data, equity research, and industry interviews.¶ The model is designed to be realistic, but its conclusions are necessarily tentative. Projecting the returns to investments of this complexity requires making many contestable assumptions, and any given real-world project will differ in multiple ways. The model is built primarily on public data, while much of the relevant information is proprietary. The model’s value lies more in the trends it reveals than any specific figure. A more complete explanation of our methods is in the appendices.¶ Three primary findings are relevant to policymakers.¶ Finding 1: Time to power is the most important driver of economic returns.¶ **Delays in getting a data center up and running are extraordinarily costly.** Our model shows that for **a typical 100 MW U.S. data center,** each additional year of delay costs roughly \$550 million in life cycle value, about 5.5 percent of the data center’s **roughly \$10 billion life cycle value** (see figure 3).3¶ Of all the external variables we considered—construction and labor costs, tariffs, local taxes, power costs, efficiency losses from the climate—delays are the number one driver of lost returns in our model, by a large margin.¶ In the United States, our model suggests that a one-year delay costs about 25 percent more than a doubling in electricity prices from their current level (\$441 million), about 60 percent more than losing typical state tax incentives (\$338 million), and three times as much as moderate tariffs on servers (\$172 million). Even the cost of a three-month delay—about \$178 million—greatly exceeds the impact of removing upfront bonus tax depreciation (\$61 million) or modest tariffs (\$43 million) (see table 1).4¶ Why Do Delays Matter So Much?¶ Three factors dominate:¶ **Delayed revenue. Every month a data center is delayed is a month of GPU-hours that could have been sold.** At current prices and utilization rates, a 100 MW Blackwell facility generates nearly \$200 million in monthly revenue as it starts operating at full capacity.5 Delays push those revenues into the future, and **at any reasonable discount rate, earlier cash flows are worth substantially more** than later ones.¶ Idle capital. Capital allocated to a planned data center earns nothing. Today, little funding may be tied up in some delayed projects, as the most expensive elements of a data center, the chips, are typically delivered at the end of the construction process. Even so, some capital will be committed in construction or labor before chips arrive. And for projects financed through debt or other mechanisms, **funding often has to be secured in advance, meaning that delays,** which lead to compounding interest, can **cause deals to be renegotiated or to fall apart altogether.**6¶ In the future, these dynamics may worsen as operators lock in longer-term contracts with chip designers or produce chips in house. Delivery schedules may become fixed in advance, in which case delays could cause chips to sit in warehouses if power isn’t available to run them.¶ Compounding effects. The faster companies can bring data centers online, the more computational power they can devote to training their AI models and researching future breakthroughs. When the data center owner or user is a model developer, this confers a critical advantage **in the AI race. Stronger models mean more customers, more revenue,** and better internal tools to help produce the next generation of models. As end-users move from pilots to wider implementation of AI systems, **leading in AI** over the next few years could **prove immensely profitable.**¶ The result is that time swamps every other factor. For the world’s latest and largest data centers, getting online months earlier is worth billions of dollars. **Policymakers focused on energy prices or construction costs are missing** a crucial part of the picture. **The decisive variable is speed.**¶ Where Delays Are Worst¶ Delays are widespread, but our research finds substantial variation across countries in time to first operation—how long it takes to bring a data center from project approval to running chips (see figure 4).7¶ Each estimate comes with high uncertainty, as projects vary within each country and each estimate accounts for both projects that connect to the grid and those that use behind-the-meter power. But on average—before accounting for possible delays from the war in Iran—the UAE leads at around twenty-two months.8 The United States follows at twenty-four months—competitive, but largely in favorable regulatory environments like Texas, not California.9 Most countries fall in the two-to-three-year range, some aided by government policies to expedite approvals. France, for example, has identified a number of turnkey sites for large-scale AI data centers. Typical projects in the United Kingdom take an estimated thirty-four months, with notable differences in grid connection times between Scotland, where timelines are roughly three years, and the rest of the country, where they can stretch to five or more. (The model’s estimated timelines are faster than typical grid **timelines,** as they also consider more rapid behind-the-meter approaches.) South Korea and **Germany** round out the pack **at** forty and **forty-two months** respectively. The **German** grid is one of Europe’s most congested—some **developers are** reportedly **turning to Belgium because of** the backlog—and South Korean grid improvements are often **delayed.** 10¶ These figures, which are derived from completed and announced projects, represent roughly median estimates for time to power for a major data center operator in each country. They are supplemented by additional research on grid and regulatory environments.

They exclude initial pre-construction land permitting and planning (which are typically nonpublic, difficult to measure consistently across countries, and often contracted to third parties).¹¹ We discuss additional methodology details in appendix 2. Actual timelines vary widely by site, jurisdiction, and execution. But while specific projects will move faster or slower than these figures, the relative ordering is meaningful.[¶] The gap between the fastest and slowest is striking: Our data suggests that the same data center could be built almost eighteen months faster in the United States than in Germany.¹² And even among democracies, the spread is substantial. OpenAI energized the first phase of its Stargate Abilene data center in Texas, with 147 MW of power, in thirteen months. In contrast, it took the American company Colt more than four times as long (six years) to begin operating a smaller data center (100 MW) in Mumbai, India, after purchasing the land.[¶] Finding 2: The United States and the UAE are the most competitive of the countries studied, while Germany faces the biggest hurdles. Changes in timelines reshuffle the relative rankings dramatically.[¶] **Because time matters so much, small changes in project timelines can reshuffle country rankings.**

The model's estimates of relative returns closely track the speed at which each country can get projects up and running (see figure 5). As a result, **better policy has the potential to vault countries** that currently perform poorly **up** the table.[¶] In our default scenario, the United States ranks second among the ten countries studied in terms of data center life cycle value. But that ranking is vulnerable.

If the typical project speeds up by nine months, the United States overtakes the UAE **as the most competitive site for AI data centers** (see table 2). **Add a one-year delay and the United States falls to fifth**— behind the UAE, Finland, India, and Canada.¹³[¶] This sensitivity explains why companies are so focused on time to power. As one Wall Street analyst told us, “power rules everything right now. If you have power that you can plug a GPU into, you’re going to get demand, whatever the power costs.”¹⁴ **An executive choosing between building in Texas or India is generally less focused on electricity prices than on how many months sooner the Texas facility can start generating revenue.** A few months’ difference can be worth hundreds of millions of dollars for a 100 MW data center.[¶] As a result, the UAE’s faster general timeline translates into a substantial valuation premium. If **India** could build as

quickly as the UAE, our model suggests it **would overtake the United States**, Finland, and Canada, despite the cheap power and the cool climates in the latter two countries, which increase data center efficiency.[¶] While the UAE leads in our results, the baseline scenario does not consider the impact of the recent U.S.-Iranian war and ongoing security risks in the region. The war may raise the difficulty of attracting the skilled workforce necessary for large projects—power grid engineers; construction managers; and civil, mechanical, and electrical engineers. Some data center developers in the Middle East have changed in-person work requirements due to safety risks, and one workforce analyst suggests that salaries may include additional hazard pay going forward. The war has also disrupted key Emirati ports like Jebel Ali and Fujairah, making it more difficult to receive raw materials needed for AI data center construction.[¶] The war also poses financial risks: mounting capital expenditures for physical data center fortifications, the need for redundancy, tighter constraints on securing capital, and higher insurance costs. Many traditional commercial insurance policies do not cover risks from armed conflict, such as an AWS data center being struck by an Iranian Shahed drone. According to one estimate, the cost of Gulf “war insurance” surged nineteen-fold to \$5 million for \$100 million in coverage after the outbreak of hostilities, with insurers offering reduced coverage limits.¹⁵ While some macroeconomic measures of risk have stabilized since the outbreak of hostilities, the Iranian military’s public threats against AI infrastructure, including OpenAI’s Stargate UAE data center, cast a long shadow over planned AI expansions in the Middle East.[¶] Although our baseline scenario does not factor in the impact of the war, we consider here some additional scenarios to illustrate how the conflict might affect projects in the Gulf. The resulting estimates are highly uncertain, as good data on the war’s effects is hard to come by, but they give a general guide to the potential impact of the conflict. The upshot is that a short war will likely do only limited damage to the UAE’s competitiveness, but if the conflict drags on, things will get much worse. In the first scenario, we assume the war delays construction timelines by six months, shipping disruptions increase non-IT construction costs by 10 percent, and the annual premium for war insurance is 0.5 percent of capital expenditures. Those **changes push overall economic returns**

in the UAE to about 1.7 percent below those in the United States. In a prolonged war scenario, where conflict delays construction by a year, non-IT construction costs rise 20 percent, and insurers charge 1 percent of capital expenditures in annual war risk premiums, the UAE would finish in sixth place, about 7.8 percent behind the United States (see figure 6).[¶] Finding 3: Tariffs, electricity prices, tax incentives, and non-IT costs also matter.[¶] Time, important as it is, isn’t the only factor. Many projects are being planned and built in places with longer than average timelines. In projecting financial returns, several other factors are also important.[¶] Tariffs. Moderate tariffs on GPU servers—a 20 percent tariff on Taiwan-assembled servers with 60 percent pass-through to buyers—would cost \$172 million in life cycle value for a U.S. data center. Even modest tariffs (5 percent with the same pass-through) would cost \$43 million. With trade tensions ongoing and semiconductor supply chains under scrutiny, tariff scenarios are plausible enough to take seriously.[¶] Electricity prices. A 50 percent increase in U.S. electricity costs would reduce life cycle value by \$221 million.¹⁶ This matters, but less than many assume. In part because electricity is a smaller share of total costs than often believed. In the United States, electricity costs are generally less than 10 percent of annual revenue for AI data centers, and less than 5 percent under the baseline assumptions of our model (see figure 7). Even with more aggressive assumptions about power costs, the investment firm New Street Research reckons that depreciation and amortization of IT equipment (chips and servers, which are replaced every three to six years) is about seven times as costly as electricity on an annual basis.[¶] Tax treatment. Removing estimated U.S. state AI data center tax incentives costs around \$338 million. Removing 100 percent bonus depreciation for equipment—the ability to deduct equipment costs upfront rather than over several years, which was made permanent under the One Big Beautiful Bill Act—costs \$61 million.[¶] The hierarchy is clear: delays have the largest impact, then electricity, then taxes and server tariffs, then other factors.[¶] Other factors. Doubling turbine capital expenditures (relevant for behind-the-meter scenarios) costs \$368 million. Higher cooling costs in hot climates, captured in the model as PUE (power usage effectiveness) losses, cost \$60 million compared to a cool-climate baseline like Finland.[¶] The hierarchy is clear: delays have the largest impact, then electricity, then taxes and server tariffs, then other factors.¹⁷ Policymakers working on data center competitiveness should allocate their attention accordingly.[¶] Why Non-IT Costs Matter Less Than You’d Think.[¶] One reason time dominates is that most data center capital expenditure goes to IT equipment—primarily GPUs—which is priced similarly around the world, rather than to construction, electrical systems, or cooling, where costs differ more across countries (see figure 6).[¶] According to research by Bank of America, IT equipment accounts for about 80 percent of capital expenditures in current data centers. Other upfront costs—which include engineering, construction, electrical systems, thermal equipment, and generators—account for only about 20 percent.[¶] This has an important implication: factors that affect non-IT costs (construction labor, permitting fees, land prices) have a relatively small impact on overall project economics. For example, despite substantially lower construction and labor costs, India’s slower timelines cause it to finish behind the United States.[¶] Countries can differentiate themselves primarily through factors that affect time (permitting, grid connections), operating costs (electricity prices, taxes), or demand (market access, data localization requirements). Of the supply side factors, time is the most important lever policymakers can pull.[¶] Although our model focuses on supply-side factors, demand is the most important ultimate driver of returns (see figure 9). A 5 percent increase in steady-state revenue—reflecting stronger AI demand or better pricing power—boosts life cycle value by about 8.5 percent, vastly exceeding the UAE’s 3.5 percent location advantage over the United States.¹⁹ Policymakers should remember that whatever levers they can pull to enable data center development, the ultimate success of the industry will depend on customer demand, and they should pause before offering subsidies for already well-financed projects.²⁰ [¶] It is also worth noting that revenue expectations in turn affect the relative importance of different policy options. When data centers are highly profitable, tax rates matter more, as there are more profits to tax. When data centers are less profitable, operating margins matter more, increasing the importance of power costs.²¹ If profits are high enough, all costs become less significant, and fast access to a large power supply dominates site selection.[¶] Understanding Time to Power.[¶] The importance of time to power makes it worth understanding what drives delays. The biggest problem for most operators is hooking up their projects to the grid. Here, much of the blame falls on regulatory delays and equipment shortages. Some companies are turning to self-help: setting up their own sources of power on site. This can speed things up but is at best a partial solution.[¶] The Grid Connection Problem.[¶] Most operators want to connect to the grid because large AI data centers consume enormous amounts of power. One gigawatt is the entire output of a typical nuclear power plant; a 5 GW cluster will use more power than London on average.²² Yet most grids simply don’t have that kind of spare capacity and aren’t adding it fast enough to keep up with demand.[¶] That is because, in many countries, getting new power sources online now takes far longer than it used to. In the United States, for example, it took an average of five years to connect a new source to the grid in 2023, up from less than two years between 2000 and 2007. In 2024, new generation projects bringing more than 20 MW to the grid faced a median wait time of thirty months just to get their request approved; beginning commercial operations took another thirty-five months, up from just eleven months for projects completed in 2007.[¶] This is in part because the nature of electricity generation has changed. Renewable energy projects—which tend to be smaller and more numerous than traditional power plants—have overwhelmed the system for connecting new power sources to the grid, a process designed for a different era.[¶] Equipment shortages are another problem. Electricity flows most efficiently over long distances at much higher voltages than consumers can use. To switch electrical currents between the two voltage levels, the grid uses electrical transformers. These pieces of equipment are essential to extending the grid, but in recent years they have been in short supply. Wait times for large transformers—most of which are imported—now average more than two years in North America and can stretch up to five, in some cases, in Europe. Large transformers also need to be custom-made, and a shortage of skilled labor, a cyclical industry, and tariff uncertainty (many critical parts are made in China) have further increased lead times.²³ [¶] The problem is even worse in some other countries. In India, one researcher writes that “typical interstate lines take 30 to 36 months to complete,” even before accounting for right-of-way and land acquisition disputes that “routinely add years to the process.” In South Korea, more than half of power upgrade projects have been delayed in recent years. The East Coast–Capital Region transmission line project, for example, has been pushed back eight years from its intended completion date in 2019. Providers in the United Kingdom have seen wait times of ten years or more to add power to the grid, although the government has introduced recent reforms in the hope of improving things.[¶] Even when the grid has capacity to support significant new demand, the electricity still has to get to the company that wants to use it. For large new projects outside population centers—most big data centers—that means upgrading or installing new transmission lines. These are specialized pieces of equipment, and companies often face long wait times. From 2017 to 2021, the United States built an average of 700 miles of new transmission lines, down from an average of 2,000 miles between 2012 and 2016.[¶] Such delays contrast with countries like the UAE, where officials can direct state-owned utilities to prioritize AI infrastructure. In most other countries, utilities must work through complex regulatory processes, often including lengthy grid stability studies, interconnection queues, and competing demands. The UAE’s tight coordination across utilities allows fast addition of power infrastructure. In contrast to the United States, where such projects involve multiple utilities, regional grid operators, regulators, landowners, and

local governments.²⁴[¶] Behind the Meter Gas: A Partial Solution.[¶] **Facing years-long grid connection waits, many data center operators have turned to behind-the-meter (BTM) power—generating electricity on-site rather than drawing from the grid.**²⁵ This usually

involves setting up clusters of natural gas turbines or reciprocating engines. These approaches bypass long grid queues, but they come with downsides, too: onsite power costs more, struggles to support the largest loads, and isn’t a permanent solution for most projects.²⁶[¶] The most common approach today is to rely on natural gas. xAI’s Colossus facility in Tennessee, for example, uses roughly 450 MW worth of gas turbines.²⁷ OpenAI’s Stargate power in Texas relies on some of its initial phase. Meta’s planned facilities include significant on-site

generation capacity, including 400 MW in Ohio. BTM configurations can cut time to first operation by a year or more. In the United States, it typically takes one to two years to procure BTM power. The process isn't instant, since much of the available equipment has been back-ordered, but it still shaves significant time off grid connections.²⁸ Although the United States has led in the deployment of BTM power, the approach is gaining traction elsewhere.[¶] Despite their rapidity, BTM turbines and other sources of on-site power have several downsides that lead industry insiders to call them a "temporary solution at best." These include the following:[¶] Supply constraints. As more operators turn to BTM, equipment is getting harder to find. Industry analysts told us that delivery timelines for large turbines (those with roughly 50 MW capacity or greater) have stretched to two to three years or more. GE Vernova, one of the largest turbine manufacturers, says it has sold out its inventory through 2028. Smaller turbines remain available on shorter timelines, but even those supplies are tightening.[¶] Higher costs. On-site generation is more expensive per kilowatt-hour than grid electricity in most markets. Fuel must be delivered; equipment must be maintained; economies of scale are harder to achieve. And reliability requirements are high for AI data centers, necessitating building overcapacity and using batteries in many cases. Our model shows that BTM turbines have higher operating costs than grid-connected scenarios, partially offsetting the time advantage. The cost is worth it to shorten timelines, but operators will likely save money if they can switch projects to grid power down the line.[¶] Environmental concerns. Gas turbines produce emissions—both carbon dioxide and local pollutants like nitrogen oxides. Google, for example, is building a gas-fired plant to power a data center in Texas that could emit more carbon dioxide by itself each year than the city of San Francisco. In addition to the direct damage it does to human health and the environment, pollution also creates political and regulatory risks, particularly in jurisdictions with aggressive climate goals. Onsite power may be preferable to keeping coal plants online, but engines and turbines still typically involve burning fossil fuels. The hyperscalers have set their own emissions reductions targets, as well, although those targets look shakier the scale of the companies' AI ambitions.[¶] Scale limitations. BTM works well for facilities in the hundreds of megawatts. But the largest planned data centers will draw gigawatts of power, the equivalent of a medium-sized city. Powering the largest data centers with on-site gas turbines will be difficult, thanks to reliability issues, contractors' lack of experience with BTM power at this scale, and regulatory uncertainty. At some point, grid connection will be preferable for most projects.[¶] As a result, most operators view BTM as a bridge while they wait for grid connections or new generation capacity to come online, not as a permanent power strategy. The long-term economics favor switching to grid power in most markets.[¶] Despite these limitations, BTM has been essential to maintaining U.S. competitiveness in AI infrastructure. With fifty states, the United States has many different state regulations on BTM power, many of them favorable toward data center developers. Without the ability to

bypass grid queues, American data center timelines would look more like Germany's.[¶] II. The Geopolitics of Compute[¶] Compute is essential to transformative AI. The countries that host compute can tax it,

control who has access to it, and regulate what they do with it. If democracies lead the development of AI, they can shape it for the better.[¶] Technological revolutions have always elevated new resources to strategic importance. The internal combustion engine turned oil into a defining resource of the twentieth century; railroads

did the same for steel and coal a generation earlier.[¶] AI is producing a similar shift. AI systems will soon drive economic growth, create new military capabilities, and accelerate scientific and industrial research. Countries that do not adopt and deploy them at scale risk seeing their economic output, institutional capacity, and geopolitical leverage atrophy compared to those that do.[¶] Why Domestic Compute Matters[¶] AI is built on talent, data, and computing power, but of these, compute is the input most amenable to deliberate policy action. It is the fastest to amass: A data center begins generating value the moment it comes online, while talent pipelines and data advantages take years to build. It is a bounded physical resource, so its location and ownership can be tracked and controlled in ways that data cannot. And it sits at a chokepoint of concentrated suppliers—ASML, TSMC, Nvidia—where policy interventions can shape who has access. As researchers at Tsinghua University have put it, chips are "the one and only physical basis" for AI development. "No chip, no AI."[¶] Countries that host AI data centers capture direct economic value: construction spending, operational jobs, tax revenue—the last of which may be essential to public finances if AI reshapes economic activity. But equally significant is what compute enables downstream. Firms with low-latency access to abundant compute can deploy AI systems that boost productivity across sectors, from drug discovery to logistics to software development. The most compute-intensive work—training models, running large-scale simulations, accelerating scientific discovery—requires infrastructure that only a handful of countries can currently provide, and hosting it may help attract the researchers and engineers who push the field forward. Over time, this can produce a virtuous cycle: infrastructure investment attracts human capital, which generates breakthroughs, which justifies further investment.[¶] Of course, wherever the compute is located, it still matters who owns it. A data center built by Microsoft in Kenya or by OpenAI's cloud partners in the United Kingdom is still largely foreign-operated infrastructure. The host country does not automatically control the models trained there, and the underlying hardware, such as advanced chips and networking equipment, will remain reliant on foreign supply chains. AI systems also require frequent updates to remain competitive, meaning any version running on a local data center will fall behind without ongoing access to the provider.[¶] Still, physical presence has advantages that remote access does not. A government that hosts compute on its soil can require cloud providers to work with local partners, enforce data residency rules, tax the data center's operators, and encourage knowledge transfer. It is easier to regulate, inspect, and negotiate with an entity that has sunk capital into your jurisdiction than one serving you remotely from another continent.[¶] These dynamics play out across several domains.[¶] Military and intelligence. A country that can train and run defense-relevant AI models on home soil retains control over the models, the data they process, and the operational security of the entire pipeline. Without domestic compute, a state's military and intelligence capabilities could become dependent on foreign providers. In a crisis, access might be throttled or cut when it is needed most.²⁹[¶] Governance. States with significant domestic compute are far better positioned to shape international standards on AI safety, military use, and other dimensions of governance than ones that depend entirely on others for access to the underlying

infrastructure. While the countries that are home to the most advanced model developers have the first claim on setting the rules, those that provide the infrastructure on which the model is trained and run will also have a substantial say

over what safety evaluations are required, what values are embedded, and what uses are permitted or prohibited.[¶] Resilience. Dependence on foreign compute creates vulnerabilities beyond the military domain. AI capabilities cannot be stockpiled, and countries that rely on remote providers face the possibility that access could be restricted—whether because exporters want to squeeze prices or redirect limited computing capacity toward other priorities, or as a result of sanctions, export controls, corporate policy changes, or external physical or cyberattacks. As one analyst has put it, "Being a net importer of a core commodity in the modern economy is seldom an excellent place to be."[¶] Fiscal stability. If AI-driven automation shifts a substantial share of economic output from labor income to corporate income, countries that host none of the relevant corporate activity may see their tax base erode with no corresponding new revenue stream. Locating as much frontier AI activity as possible within a country's tax jurisdiction—through data centers, local subsidiaries, or negotiated revenue-sharing arrangements—may provide a pathway to fiscal stability that pure importers of AI services will lack.[¶] A country or coordinated bloc that hosts a meaningful share of global AI infrastructure could gain something analogous to what OPEC members hold in energy markets. [¶] A country or coordinated bloc that hosts a meaningful share of global AI infrastructure could also gain something analogous to what OPEC members hold in energy markets. Even a single-digit share of global supply, if it can credibly be taken offline, would likely be enough to inflict real costs. Onshore compute does not guarantee sovereignty over AI. But it does make it far more likely that a country can influence how AI develops.[¶] The Balance of Power[¶] The distribution of compute will shape not only the power of individual nations, but also the contest between the United States and China and the standing of democracy on the global stage.[¶] The U.S.-China competition. For now, the United States has a significant lead in compute: In 2025, an estimated three-quarters of the world's advanced AI computing clusters were on American soil, with just 14 percent in China. China likely reached only 1–4 percent of U.S. AI chip production in 2025 and will hit only 1–2 percent in 2026. But China has launched an all-out effort to improve its position across the AI supply chain, pumping more than \$150 billion into the industry since 2014, supplemented in 2024 by an additional support package worth as much as \$47.5 billion. The Chinese system can override local opposition, fast-track permitting, and direct capital with a unity of purpose that democracies rarely match. China also enjoys advantages in key inputs: it produces more STEM PhDs than the United States each year, and its electrical generation has soared from one-third of U.S. levels in 2000 to nearly two-and-a-half times U.S. levels in 2024 (see figure 11).³⁰[¶] U.S. export controls have constrained Chinese access to the most advanced chips, but Chinese capabilities are advancing and the country has more than enough power and land to run every chip it can plausibly produce. If democratic buildouts

slow while China accelerates, the balance of AI capability—and the broader balance of economic and military power that will increasingly track it—will shift.[¶] The U.S.-China rivalry is the sharpest expression of a broader dynamic. Authoritarian governments argue with growing confidence that their model of governance delivers results—that centralized decisionmaking, state-directed investment, and the ability to override dissent produce faster growth and more effective responses to national challenges. If democratic states cannot maintain their lead in AI, they will reinforce the narrative that democratic governance is too slow and too fractious to compete in an era of rapid technological change.[¶] If democratic states cannot maintain their lead in AI, they will reinforce the narrative that democratic governance is too slow and too fractious to compete in an era of rapid technological change.[¶] AI also poses an even deeper threat to democratic governance. Historically, authoritarian regimes have faced inherent constraints that eventually checked their power: Economic growth requires skilled workers who may demand political rights, and effective militaries risk becoming independent power centers. Advanced AI could loosen both. Automated systems could reduce dependence on workers who might organize for reform, while AI-enabled surveillance lowers the cost of monitoring and suppressing

dissent. Many authoritarian regimes, including China, currently spend as much on domestic security as they do on external defense; AI could allow that control to scale more efficiently, entrenching authoritarian rule and extending its reach abroad.¶ There is also reason to think **authoritarian AI development increases the risk of catastrophic outcomes.** The **conditions that produced** disasters like **Chernobyl** and the destruction of the Aral Sea — **lack of transparency, suppression of dissent, centralized target-setting, reluctance to report bad news**—are precisely the **conditions that AI safety researchers warn against** in the development of powerful systems.³¹ As Turing Award winner Yoshua Bengio has argued, **authoritarian regimes are “more likely to make unsafe decisions regarding powerful AI systems, thereby increasing the likelihood of catastrophic outcomes.”**¶ The United States and its allies cannot prevent authoritarian states from developing powerful AI indefinitely. But **the tighter the race, the greater the pressure on democracies to cut corners** on safety. A clear **democratic lead reduces** that **pressure** and increases the likelihood that frontier AI is developed under **conditions that allow for genuine oversight.** Democracies do not need a monopoly on AI. But they do need to stay ahead.¶ **Open societies** have generally **outperformed closed ones** because they better harness human capital and **innovation.** Democratic countries need to pull off that trick again by figuring out how to develop and deploy beneficial AI before their competitors do. None of this means democracies should ignore legitimate concerns about land use, energy consumption, and corporate influence. But it does underscore the stakes: The cumulative result of decisions around permitting reform and grid connections will shape the contest between political systems in the twenty-first century.

Aff AI regulations backfire.

Rodskog 26 [Thomas Rodskog (Founder of AI Safety @ UCSB, Student @ UCSB), 03-15-2026, "Sanders's Data Center Moratorium Is Risky Strategy for AI Safety", LessWrong, <https://www.lesswrong.com/posts/GSD8bEjREYioBDIsr/sanders-s-data-center-moratorium-is-risky-strategy-for-ai>, DOA: 7-9-2026] | prav

I, too, think these concerns are very serious. But I’m not confident that a 2026 **data center moratorium** addresses them, and I worry it might **leave us even worse off.** I think a temporary data center moratorium is unlikely to meaningfully slow AI development, and more importantly, it risks associating AI safety with weak environmental arguments and left-populist politics in a way that could generate backlash and make more important regulation harder to pass.¶ It probably won’t work.¶ Even in the best case for this legislation, a data center moratorium is a temporary measure. Without concrete political pressure for a pause on superintelligence, there are just too many forces fighting against this type of legislation. National security is, rightfully, an important concern in our political climate. AI development must continue to keep up with China’s military capabilities. The entire economy is also betting on AI-powered growth. There are **extremely well-funded actors, from the major AI labs to the venture capital firms** backing them, that **will lobby aggressively against** any construction freeze, and **they have far more political leverage than Sanders** does. Without broad public support, which is building but not here yet, **this thing will get repealed** (assuming it could even pass).¶ And even while a moratorium is in effect, it only targets one part of the capabilities pipeline. Existing data centers keep running, and capital will be reallocated to do better research with existing compute, or build up energy infrastructure in preparation for when data center construction can resume. The labs don’t just sit around. This means that even a full year of a construction freeze (which seems wildly unlikely given the current administration) might only delay frontier AI capabilities by a matter of months.¶ What does Bernie plan to do with that extra time?¶ The economic concerns about job loss and disempowerment don’t get magically solved with this slowdown. His efforts would be better spent working on the actual policy responses to automation and economic disruption. And on the issue of existential risk, a unilateral U.S. moratorium doesn’t slow down China. If powerful AI is going to be built, I would rather it be built by labs with alignment researchers in a country with a free press, accountable to its citizens. The real solution to AI x-risk is not a domestic construction ban. We need to be working toward a treaty banning the development of superintelligence, verified through international monitoring of data center compute, analogous to nuclear non-proliferation.¶ I don’t think this policy would be very effective. But the bigger risk isn’t that **the moratorium** fails to slow down AI. It’s that it sets back the political movement that actually could.¶ It **might backfire.**¶ “Data center moratorium” is, in the public mind, not really about existential risk. To Sanders’s credit, his announcement actually focused on the right things: existential risk, economic concerns, and political disempowerment. But the growing **public opposition** to data centers **is built on** a coalition of concerns, and much of that coalition is driven by **environmental complaints** about water usage **and energy** consumption that I don’t really buy.¶ A **bill** banning data center construction **is going to get interpreted through that lens** regardless of Sanders’s intent. **There’s a reason he proposed banning buildings instead of, say, taxing compute.** The message is simple, and it maps well onto a public that already thinks data centers are evil water-sucking entities. I don’t love the idea of **the AI safety movement** **becom**ing **associated with** arguments that are both **wrong** and

very partisan. It might be useful to ride the wave of populism, but if you're not careful, it will backfire. Just look back at the tech-right backlash to progressive overreach in 2024, which is still shaping AI policy today (see David Sacks). A data center moratorium from Bernie Sanders is exactly the kind of thing they will point to as evidence that AI regulation is just left-wing populist environmentalism dressed up in safety language. That framing makes it harder to pass serious compute governance proposals down the line, because every future attempt at regulation has to fight through the association. Stopping ASI development should not be partisan, and it doesn't need to be populist. It needs to be common sense.¶ How this could go well.¶ All that said, the current situation might still be fine. I think in an ideal scenario, the legislation moves the idea of a pause into the Overton window, but does not pass. It helps Washington wake up to the risks of advanced AI without polarizing people in tech or strongly associating AI safety with environmental issues. If that's how it plays out, great.

No US-China cooperation

Chang 26 [Chang, Gordon (J.D. from Cornell, Political Analyst), 05-12-2026, "Will Bernie Sanders Hand AI Dominance to China?", Newsweek, <https://www.newsweek.com/will-bernie-sanders-hand-ai-dominance-to-china-opinion-11930125>, DOA: 7-13-2026] | eti

Senator Bernie Sanders wants to prevent the "runaway train" of unregulated development of artificial intelligence by working with China.¶ The Vermont independent has a point, as anyone who has watched the Terminator movies knows, but cooperation with the Chinese state would nonetheless not be in America's interest.¶ Sanders is passionate about the issue. "What I don't understand is how anyone can ignore when the world's leading AI scientists tell us that, as AI becomes smarter than humans, it could operate independently of human control—with possible catastrophic results," he wrote on X on April 30.¶ The day before at the U.S. Capitol, Sanders hosted American and Chinese scientists who advocated cooperation. "If President [Donald] Trump can sit down with [Chinese] President Xi [Jinping], I think leading scientists in America can sit down with leading scientists in China to discuss how in fact we can work together," he told reporters.¶ Trump may discuss AI issues with Xi when they meet in Beijing this week. If so, this will not be the first time an American leader has tried to start a conversation on the topic.¶ President Joe Biden raised the issue of AI cooperation with Xi when the two leaders met in California in November 2023. Chinese officials in May 2024, however, rejected talks with America on limiting the use of a artificial intelligence for controlling the launch of nuclear weapons.¶ "Nobody wants to see AI-controlled nuclear weapons, right?" asked Joe Wang, a former State Department and National Security Council (NSC) staffer at the Arlington, Virginia-based Special Competitive Studies Project, which specializes in AI and emerging technologies. "Like, even the craziest dictator can probably agree."¶ Whether crazy or not, Xi is not willing to talk about the topic.¶ Perhaps that unwillingness is fortunate for the United States. Yes, no American official wants China's machines to have the ability to trigger Armageddon, but that does not mean cooperation is possible.¶ Or in America's interest. "Bernie Sanders's neo-Luddite stance on AI and his urging that the U.S. capitulate to China by acceding to CCP blandishments on AI development 'cooperation' amount to de facto calling for America's surrender in the hybrid war that Xi Jinping is waging for global supremacy" Charles Burton of the Sinopsis think tank told me this month. "Beijing must be rejoicing at Sanders's wrong-headed naivety."¶ In the race for AI supremacy—sometimes characterized as the race for the 21st century—the U.S. holds a lead. Cooperation would tend to erode that edge as water runs downhill. "Of course, Beijing is growing increasingly concerned that the Trump administration has lit the afterburners of innovation, setting the U.S. to hold the lead in AI," Blaine Holt, a retired U.S. Air Force brigadier general who comments on China and other matters for Newsmax, told me, in response to Sanders.¶ "Panic must be setting in because the Communist Party's system is not an incubator that nurtures new technologies like the U.S.," Holt added. "So China must turn to its 'friends' in D.C. to hold back America. Maybe Sanders would be so kind to describe why compromising a competitive advantage is in our national interest."¶ Sanders and Representative Alexandria Ocasio-Cortez (D-NY) unveiled their Artificial Intelligence Data Center Moratorium Act, which would prohibit the construction or upgrading of new AI data centers until Congress enacted a regulatory framework. Sanders office, as Fox News reported, said the proposed law was designed to "slow down the development of AI."¶ The Vermont senator, who has pointed to Cold War-era arms control agreements as models, wants a global pact on AI.¶ Sanders needs to pick a better example. Beijing has repeatedly rejected Washington's attempts to discuss nuclear arms control, and China has—despite its obligations in the global Nuclear Non-Proliferation Treaty—spread nuclear weapons technology, equipment and materials to, among others, Pakistan, Iran and North Korea. America, on the other hand, prevented friends and allies from developing such weapons.¶ Moreover, a global AI pact would be completely unworkable. For one thing, it would be hard for a free society to ensure compliance as AI development is not something that can be monitored like warheads on top of missiles.¶ An agreement would be difficult to enforce against others as verification

would be virtually impossible for the same reason.¶ There is no question that AI, as Sanders says, could pose an “existential threat,” but an agreement with China is not the answer as there is no evidence that Xi Jinping is more concerned about the fate of the human race than that of the Chinese he rules. So, perhaps he is more dangerous than the “craziest dictator.”¶ Is there any solution?¶ Perhaps the only one is to let Americans innovate, dominate AI and trust the good instincts of their tech community to develop sensible and humane standards. Such standards are unlikely to emerge from a world dominated by the researchers of a militant Chinese Communist Party.

Xi would be incentivized to maliciously use AI--Taiwan invasion!

Agee 25 [Ryan (senior US military officer with 25 years in national security and special operations), 10-02-2025, “China’s Closing Window: Strategic Compression and the Risk of Crisis,” Foreign Policy Research Institute, <https://www.fpri.org/article/2025/10/chinas-closing-window-strategic-compression-and-the-risk-of-crisis/>, DOA: 2-27-2026] prav

Strategic Compression: A Closing Window¶ Compression is the pressure Beijing places upon itself; friction is the pressure others apply upon Beijing. Strategic compression is best understood as the simultaneous narrowing of options and acceleration of timelines that confront states when long-term ambitions collide with structural constraints. The concept is related to the classic “windows of vulnerability” debate in strategic studies. Lebow argued that **leaders exploit perceived openings when adversaries are temporarily weak**. Later scholars extended the idea: Doerer and Eidenfalk posited that foreign policy change often requires leaders to perceive a fleeting “window of opportunity” as both real and urgent. More recently, Hal Brands has tied these windows directly to great-power competition, warning that compressed states may lunge an advantage before their window of opportunity closes.¶ **China today is a textbook case. Its demographic collapse** (fertility below 1.1, shrinking labor force, and surging pension obligations) **erodes both economic and military potential. Its economic stagnation** (youth unemployment above 20 percent, debt-to-gross domestic product (GDP) ratios beyond 270 percent, and **growth rates far lower than official statistics claim**) **undermines the CCP’s claim to technocratic competence**. And its political rigidity (Xi Jinping’s consolidation of power, purges of technocrats, and reliance on loyalty over capability) reduces the very adaptability compression demands.¶ Compression is both structural and perceptual, but the latter is the primary force. Leaders act not on perfect information but on how they read the trajectory of time. In Beijing, the perception is that delay is fatal: the path to “rejuvenation” risks being blocked by demographics, debt, and dissent. The 2025 white paper, China’s National Security in the New Era, makes this explicit by subordinating economic growth to security priorities and framing **Taiwan’s annexation as essential to CCP survival**.¶ Strategic compression thus creates a dilemma. Beijing can attempt reform, a path of delay, though uncertain and historically resisted. Or it can gamble that crisis, even conflict, offers the only way to break through the narrowing window. In either case, time itself has become China’s adversary.¶ CCP Attributes Under Compression¶ Demographic and Economic Decline¶ The CCP’s legitimacy has long rested on two pillars: rising prosperity and national rejuvenation. Both are faltering. China’s demographic decline is accelerating faster than even pessimistic forecasts predicted. Fertility rates have fallen below 1.1, population peaked in 2022, and the working-age cohort is shrinking by millions annually. The consequences ripple across the system: labor shortages, surging pension obligations, and a contracting pool for military recruitment.¶ Economic performance tells a similar story. While official figures continue to advertise ~5 percent GDP growth, independent estimates suggest real growth is half that. Youth unemployment (officially “suspended” from publication in 2023 after surpassing 20 percent) reflects a generation losing faith in the promise of upward mobility. Debt-to-GDP ratios above 180 percent, and likely closer to 300 percent by some estimates, reveal an economy built on fictitious capital and financial engineering. Foreign investment, once the backbone of China’s integration into global markets, has collapsed. These are not cyclical dips but structural stall points of a peaking great power.¶ Narrative Control and Crisis Politics¶ Authoritarian systems compensate for stagnation with propaganda. Xi Jinping has tightened control over media, education, and party messaging to frame China as besieged by hostile forces determined to block its rightful rise. The 2025 National Security White Paper codifies this framing, subordinating economic goals to regime security and elevating Taiwan’s annexation as a matter of existential survival.¶ This narrative thrives in crisis conditions. The CCP has demonstrated repeatedly that crises, whether real or manufactured, strengthen its grip. During the COVID-19 lockdowns, Beijing justified draconian controls as national defense of health. Maritime coercion against the Philippines or Taiwan is framed as a reactive defense against foreign encroachment. Even violent repression, from Tiananmen Square in 1989 to Hong Kong in 2019, was justified as necessary to preserve the party’s survival. The lesson Xi has drawn is that crisis validates the party; war, by contrast, carries existential risks that could expose regime fragility.¶ Gambling for Resurrection¶ Yet compression creates pressure for escalation. Downs and Roche describe this dynamic as “gambling for resurrection”: when embattled leaders, facing declining legitimacy or looming failure, take high-risk actions because the downside is already existential. Xi’s tightening personalist system makes this risk acute. At seventy-one, his own legacy timeline intersects with China’s demographic and economic timelines. Having purged rivals and eliminated dissent, he has no safety valves left.¶ This is why strategic compression is so dangerous. For a regime that thrives in crisis but not in war, the **narrowing window may incentivize exactly the kind of reckless gamble** that could prove catastrophic: a **rapid strike against Taiwan before optimal readiness**, or a manufactured external confrontation to rally internal legitimacy. These would not be rational wars of choice but desperate wars of necessity, undertaken not because success is assured, but because delay appears fatal.¶ China’s Strategic Position Summarized¶ The CCP today faces a paradox. It possesses the world’s largest navy, expanding nuclear forces, and formidable

coercive tools in the gray zone. Yet its foundations, population, economy, and governance are eroding. This creates a system both powerful and brittle: strong enough to coerce neighbors, weak enough to fear internal unraveling. In such a system, strategic compression does not just narrow options; it warps incentives. It encourages Beijing to equate inaction with defeat and to view the crisis not as a danger to avoid but as a tool to embrace.¶ US Responses to Strategic Compression¶ Strategic compression places Beijing at a dangerous crossroads, but it also creates space for Washington to shape outcomes. By applying complementary lines of effort (deterrence, strategic friction, and off-ramp architecture), the United States and its allies can deny both a war of necessity and a war of choice, holding China in stasis until its internal contradictions resolve themselves.¶ Entrapment in a limited war may look tempting, but it runs counter to the logic of compression: Beijing thrives in crisis, not in sustained conflict. A limited clash risks providing the CCP the very external confrontation it seeks to rally domestic legitimacy, tighten control, and divert attention from internal fragilities. By contrast, enforced stasis denies both the war of necessity and the war of choice, leaving China suspended between ambition and decline. The United States gains most not by provoking conflict, but by letting compression do its work; time erodes Beijing's foundations far more reliably than a limited war that could backfire into wider escalation or regime consolidation.¶ 1. Deter the War of Necessity¶ The most immediate risk is that Beijing, fearing the window closing entirely, chooses to act before it is ready. This "war of necessity" would prioritize timing over preparation: a rapid strike on Taiwan or coercive escalation elsewhere designed to achieve fait accompli before US intervention.¶ To prevent this, deterrence must make such action prohibitively costly and uncertain of success. Forward-deployed US and allied forces across the first island chain should demonstrate visible readiness and credibility. Multilateral exercises with Japan, the Philippines, and Australia should underscore that any aggression will trigger collective resistance. Revealing previously unacknowledged capabilities (hypersonic weapons, extended-range strike systems, or integrated special operations force-space-cyber concepts) can inject uncertainty into People's Liberation Army (PLA) planning.¶ Deterrence must also extend beyond the military. The precedent of SWIFT sanctions against Russia illustrates the power of coordinated economic punishment. A credible plan to cut China off from Western capital, food, or energy markets in the event of aggression would force Xi to weigh not just battlefield risks but the collapse of the regime's domestic legitimacy. The goal is not to guarantee that the PLA cannot fight, but to ensure Xi believes he cannot win.

If the US loses the AI races, the consequences are devastating.

Chambliss 25 --- (Kent Conrad Saxby Chambliss [], 12-18-2025, "America Could Lose the AI Race Unless Congress Acts Now", National Interest, <https://nationalinterest.org/blog/techland/america-could-lose-the-ai-race-unless-congress-acts-now, doa7-14-2026>) \\n\\n

To win the AI race against China, Congress must mobilize all forms of power, modernize the grid, onshore critical supply chains, and address America's STEM talent gap

America stands at a crossroads in the most consequential technology race of our lifetime. The outcome of the US-China competition for artificial intelligence (AI) will determine which country—and which set of values—leads the global economy and secures geopolitical preeminence.

While the United States was first on the AI scene, and still holds important advantages, new findings from the American Edge Project (AEP) reveal a troubling reality: we are not positioned for long-term success.

China is moving faster and with greater strategic focus across the four pillars that will decide the race: energy, compute, talent, and adoption. Unless Congress acts, America will lose its early advantage and give the Chinese Communist Party (CCP) the opening it needs to gain more ground and win this high-stakes race.

China Is Building the Foundations of AI Power Faster Than the United States

Start with energy. China is treating AI like a national industrial project—and building the power system to match it. Beijing already generates twice as much electricity as the United States and will add 60 percent more capacity by 2040. Last year, it invested \$300 billion more than

America did in energy infrastructure. Since 2009, China has built more than 18,000 miles of ultrahigh-voltage transmission lines. The United States has built zero.

Meanwhile, our own energy infrastructure is straining under modern demand. Seventy percent of US transmission lines are more than 25 years old. The American Society of Civil Engineers (ASCE) gives our energy grid a D+. And experts warn we could see a 100-fold increase in power outages by 2030 if we fail to modernize. The frustrating part is that we have the ability to rapidly scale energy generation, but we just aren't executing. Roughly 12,000 energy projects remain stuck waiting for connection approvals. Lead times for gas turbines have grown to seven years. And transformer prices have risen between 45 percent and 95 percent since 2020.

Energy is not our only gap. China is also outpacing us in talent. It is producing millions more Science, Technology, Engineering, and Mathematics (STEM) graduates each year and aggressively growing the skilled workforce needed to develop and deploy AI systems at scale. America's innovation ecosystem is the best in the world, but our talent pipeline is no longer keeping pace with global competition.

China is also moving faster on adoption. Eighty-three percent of Chinese organizations report deploying generative AI, compared with 65 percent in the United States. And **Beijing is exporting its AI ecosystem abroad, and in so doing, embedding censorship and surveillance norms into the digital infrastructure of emerging markets. If those tools become the default worldwide, global digital standards will tilt toward authoritarian control rather than democratic openness.**

Extinction

Clare 25 [Stephen Clare (former research fellow @ the Forethought Foundation) 03-xx-2025, "Great power war", 80,000 Hours, <https://80000hours.org/problem-profiles/great-power-conflict/>, accessed 08-12-25]

A modern great power war could see nuclear weapons, bioweapons, autonomous weapons, and other destructive new technologies deployed on an unprecedented scale. It would probably be the most destructive event in history, shattering our world. It could even threaten us with extinction. We've come perilously close to just this kind of catastrophe before. On October 27, 1962 — near the peak of the Cuban Missile Crisis — an American U-2 reconnaissance plane set out on a routine mission to the Arctic to collect data on Soviet nuclear tests. But, while flying near the North Pole, with the stars obscured by the northern lights, the pilot made a navigation error and strayed into Soviet airspace. Soviet commanders sent fighter jets to intercept the American plane. The jets were picked up by American radar operators and nuclear-armed F-102 fighters took off to protect the U-2. Fortunately, the reconnaissance pilot realised his error with enough time to correct course before the Soviet and American fighters met. But the intrusion enraged Soviet Premier Nikita Khrushchev, who was already on high alert amidst the crisis in Cuba. "What is this, a provocation?" Khrushchev wrote to US President John F. Kennedy. "One of your planes violates our frontier during this anxious time when everything has been put into combat readiness." If the U-2's path had strayed further west, or the Soviet fighters had been fast enough to intercept it, this incident could have played out quite differently. Both the United States and the USSR had thousands of nuclear missiles ready to fire. Instead of a nearly-forgotten anecdote, the U-2 incident could have been a trigger for war, like the assassination of Franz Ferdinand. Competition among the world's most powerful countries shapes our world today. And whether it's through future incidents like the lost U-2, or something else entirely, it's plausible that it could escalate and lead to a major, devastating war. Is there anything you can do to help avoid such a terrible outcome? It is, of course, difficult to imagine how any one individual can hope to influence such world-historical events. Even the most powerful world leaders often fail to predict the global consequences of their decisions. But I think the likelihood and severity of great power war makes this among the most pressing problems of our time — and that some solutions could be impactful enough that working on them may be one of the highest-impact things to do with your career. By taking action, I think we can create a future where the threat of great power war is a distant memory rather than an ever-present danger. Summary **Economic growth and technological progress have bolstered the arsenals of the world's most powerful countries.** That means the next war between them could be far worse than World War II, the deadliest conflict humanity has yet experienced. Could such a war actually occur? We can't rule out the possibility. **Technical accidents or diplomatic misunderstandings could spark a conflict that quickly escalates. Or international tension could cause leaders to decide they're better off fighting than negotiating.**

It seems hard to make progress on this problem. It's also less neglected than some of the problems that we think are most pressing. There are certain issues, like making nuclear weapons or military artificial intelligence systems safer, which seem promising — although it may be more impactful to work on reducing risks from AI, bioweapons or nuclear weapons directly. You might also be able to reduce the chances of misunderstandings and miscalculations by developing expertise in one of the most important bilateral relationships (such as that between the United States and China). Finally, by making conflict less likely, reducing competitive pressures on the development of dangerous technology, and improving international cooperation, you might be helping to reduce other risks, like the chance of future pandemics. Our overall view: **Recommended** Working on this issue seems to be among the best ways of improving the long-term future we know of, but all else equal, we think it's less pressing than our highest priority areas (primarily because it seems less neglected and harder to solve). **Scale** There's a significant chance that a new great power war occurs this century. Although the world's most powerful countries haven't fought directly since World War II, war has been a constant throughout human history. There have been numerous close calls, and several issues could cause diplomatic disputes in the years to come. These considerations, along with forecasts and statistical models, lead me to think there's about a one-in-three chance that a new great power war breaks out in roughly the next 30 years. Few wars cause more than a million casualties and the next great power war would probably be smaller than that. However, there's some chance it could escalate massively. Today the great powers have much larger economies, more powerful weapons, and bigger military budgets than they did in the past. An all-out war could kill for more people than even World War II, the worst war we've yet experienced. Could it become an existentially threatening war — one that could cause human extinction or significantly damage the prospects of the long-term future? It's very difficult to say. But my best current guess is that the chance of an existential catastrophe due to war in the next century is somewhere between 0.05% and 2%. **Neglectedness** War is a lot less neglected than some of our other top problems. There are thousands of people in governments, think tanks, and universities already working on this problem. But some solutions or approaches remain neglected. One particularly promising approach is to develop expertise at the intersection of international conflict and another of our top problems, experts who understand both geopolitical dynamics and risks from advanced artificial intelligence, for example, are sorely needed. **Solvability** Reducing the risk of great power war seems very difficult. But there are specific technical problems that can be solved to make weapons systems safer or less likely to trigger catastrophic outcomes. And in the best case, working on this problem can have a leverage effect, making the development of several dangerous technologies safer by improving international cooperation and making them less likely to be deployed in war. At the end of this profile, I suggest five issues which I'd be particularly excited to see people work on. These are: **Developing expertise in the closest bilateral relationships** Learning how to manage international crises quickly and effectively and ensuring the systems to do so are properly maintained **Doing research to improve particularly important foreign policies, the strategies for sanctions and deterrence** Improving how nuclear weapons and other weapons of mass destruction are governed at the international level **Improving how such weapons are controlled at the national level** Why might preventing great power war be an especially pressing problem? A modern great power war — as all-out conflict between the world's most powerful countries — could be the worst thing to ever happen to humanity. Historically, such wars have been exceptionally destructive. Sixty-six million people died in World War II, likely the deadliest catastrophe humanity has experienced so far. Since World War II, the global population and world economy have continued to grow, nuclear weapons have proliferated, and military technology has continued to advance. This means the next world war could be even worse, just as World War II was much deadlier than World War I. It's not guaranteed that such a war will break out. And if it does, it may not escalate to such a terrible extent. But the chance can't be ignored. In fact, there are reasons to

think that the odds of World War II breaking out this century are worryingly high. A modern great power war would be devastating for people alive today, but its effects could also persist long into the future. That's because there is a substantial chance that this century proves to be particularly important. Technologies with the potential to cause a global catastrophe or radically reshape society are likely to be invented. Hence we chose to develop and deploy them could impact huge numbers of our descendants. And these choices would be affected by the outcomes of a major war. It is more specific, there are three main ways great power conflict could affect the long term future. High international tension could increase other risks. Great power tensions could make the world more dangerous even if they don't lead to war. During the Cold War, for example, the United States and the USSR never came into direct conflict but instead in bioweapons research and built up nuclear arsenals. This dynamic could return, with tension between great powers fueling races to develop and build new weapons, raising the risk of a disaster even before shots are fired. It could also cause an existential catastrophe. If war does break out, it could escalate dramatically, with modern weapons (nuclear weapons, bioweapons, autonomous weapons, or other future technologies) deployed at unprecedented scale. The resulting destruction could irreparably damage humanity's prospects. War could reshape international institutions and power balances. While such a catastrophe was possible, it seems extremely unlikely. But even a less deadly war, such as another conflict on the scale of World War I, could have very long-lasting effects. For example, it could reshape international institutions and the global balance of power. In a post-war era, several institutional arrangements and geopolitical balances could cause humanity to follow different long-term trajectories. The rest of the profile explores exactly how pressing a problem great power conflict is to humanity. Great power relations have become more tense. More partly as a result, a war is more likely than you might think. It's reasonable to put the probability of such a conflict in the coming decades somewhere between 10% and 50%. If war breaks out, it would probably be hard to control escalation. The chance that it would become large enough to be an existential risk cannot be dismissed. This makes great power war one of the biggest threats our species currently faces. It seems hard to make progress on solving such a difficult problem (more) – but there are many things you can try if you want to help join. International tension has risen and makes other problems worse. Imagine we had a thermometer-like device which, instead of measuring temperature, measured the level of international tension. This 'tension metre' would max out during periods of all-out global war, like World War I. And it would be relatively low when the great powers were peaceful and cooperative. For much of the post Napoleonic 1800s, for example, the powerful European nations instituted the Concert of Europe and mostly upheld a continental peace. The years following the fall of the USSR also seem like a time of relative calm, when the tension metre would have been quite low. How much more worried would you be about the coming decades if you knew the tension metre would be very high than if you knew it would be low? Probably quite a lot. In the worst case, of course, the great powers could come into direct conflict. But even if it doesn't lead to war, a high level of tension between great powers could accelerate the development of new strategic technologies, make it harder to solve global problems like climate change, and undermine international institutions. During the Cold War, for instance, the United States and USSR avoided coming into direct conflict. But the tension metre would still have been pretty high. This led to some dangerous events. A nuclear arms race. The number of nuclear warheads in the world grew from just 300 in 1950 to over 64,000 in 1986. The development of new bioweapons. Despite signing the Biological Weapons Convention in 1972, the search for military advantages motivated Soviet decision makers to continue investing in bioweapon development for decades. Although never used in combat, biological agents were accidentally released from research facilities, resulting in dozens of deaths and threatening to cause a pandemic. Nuclear close calls. Military accidents and false alarms happened regularly, and top decision makers were more likely to interpret these events hostilely when tensions were high. On several occasions it seems the decision about whether or not to start a nuclear war came down to individuals acting under stress and with limited time. This makes international tension an existential risk factor. It's connected to a number of other problems, which means reducing the level of international tension would lower the total amount of existential risk we face. The level of tension today. Recently, international tension seems to have once again been rising. To highlight some of the most salient examples. China-United States relations have deteriorated, leading to harsh diplomatic rhetoric and protectionist trade policies that aim to reduce the countries' economic interdependence. Russia's invasion of Ukraine has killed about a hundred thousand people so far, raised the risk of nuclear war, and sent United States-Russia relations to their lowest point since the Cold War. Chinese and Indian soldiers fought deadly skirmishes along their countries' disputed border in 2020-21. These dynamics raise an important question: how much more dangerous is the world going this higher tension than would be in a world of low tension. I think the answer is quite a bit more dangerous – for several reasons. First, international tension seems likely to make technological progress more dangerous. There's a good chance that, in the coming decades, humanity will make some major technological breakthroughs. We've discussed, for example, why one might worry about the effects of advanced artificial intelligence systems or biotechnology. The level of tension could strongly affect how these technologies are developed and governed. Tense relations could, for example, cause countries to neglect safety concerns in order to develop technology faster. Second, great power tensions will strongly influence how nations do, or do not, cooperate to solve other global collective action problems. For example, in 2022, China withdrew from bilateral negotiations with the United States over climate action in protest of what it perceived as American diplomatic aggression in Taiwan. That same year, efforts to strengthen the Biological Weapons Convention were reportedly hampered by the Russian delegation after their country's invasion of Ukraine raised tensions with the United States and other western countries. And third, if relations deteriorate severely, the great powers could fight a war. How likely is a war? Wars are destructive and risky for all countries involved. Modern weapons, especially nuclear warheads, make starting a great power war today seem like a suicidal undertaking. But factors like the prevalence of war throughout history, the chance that leaders make mistakes, conflicting ideologies, and commitment problems, make me think that conflict could break out anyway. On balance, I think such an event is somewhat unlikely but hardly unthinkable. To quantify this I put the chance we experience some kind of war between great powers before 2050 at about one in three. One reason to think a war is quite likely is that such conflicts have been common in the past. Over the past 500 years, about two great power wars have occurred per century. Namely, this would mean that every year there's a 2% chance such a war occurs, implying the chance of experiencing at least one great power war in the next 80 years – roughly until the end of the century – is about 82%. This is a very simple model. In reality, the risk is not constant over time and independent across years. But it shows that if past trends simply continue, the outcome is likely to be very bad. The great power war we become too likely. One of the most important criticisms of this model is that it assumes the risk is constant over time. Some researchers have argued instead that, especially since the end of World War II, major conflicts have become much less likely due to nuclear deterrence. Nuclear weapons are so powerful and destructive that it's just too costly for nuclear armed countries to start wars against each other. Democratization: Democracies have almost never gone to war against each other, perhaps because democracies are more interconnected and their leaders are under more public pressure to peacefully resolve disputes with each other. The proportion of conflicts that are democratic has increased from around 10% in 1940 to about 50% today. Strong economic growth and global trade. Global economic growth accelerated following World War I and the value of global exports grew by a factor of almost 30 between 1950 and 2014. Since war disrupts economies and international trade, strong growth raises the costs of fighting. The spread of international institutions. Multilateral bodies like the United Nations General Assembly and Security Council promote diplomatic dialogue and facilitate coordination to punish transgressions. It is true that we are living through an unusually long period of great power peace. It's been about 80 years since World War I. We just saw that a simple model using the historical frequency of great power wars suggests there was only a 20% chance of going that long without at least one more war breaking out. This is some evidence in favour of the idea that we have become significantly less likely to fight. At the same time, we shouldn't feel too optimistic. The numerous civil wars during the Cold War suggest we were somewhat lucky to avoid a major war in that time. And a 20% chance of observing 80 years of peace is not low. Structural changes might have dramatically reduced the likelihood of war. Or perhaps we've just been lucky. It could even be that technological advances have made war less likely to break out, but more deadly when it occurs, leaving the overall effect on the level of risk ambiguous. It hasn't been long enough to support a decisive view. To solve the recent historical trend is somewhat encouraging, we don't have nearly enough data to be confident that great power war is a thing of the past. To better predict the likelihood of future conflict, we should also consider distinctive features of our modern world. One might think that a modern great power war would simply be so destructive that no state leader would ever choose to start one. And some researchers do think that the destruction such a war would wreak globally makes it less likely to occur. But it would be hard to find anyone who claims this dynamic has even reduced the risk to zero. First, a war could be started by accident. Second, sometimes even prudent leaders may struggle to avoid a slide towards war. We could blunder into war. An accidental war can occur if one side mistakes some event as an aggressive action by an adversary. This happened several times during the Cold War. The earlier example of the suspected American reconnaissance plane shows how routine military exercises carry some escalation risk. Similarly, throughout history, nervous pilots and captains have caused serious incidents by attacking civilian ships and ships. Nuclear weapons allow for massive retaliatory strikes to be launched quickly – potentially too quickly to allow for such situations to be explained and de-escalated. It is perhaps more likely that there can be an accident war could be triggered by a technological malfunction. Faulty computers and satellites have previously triggered nuclear close calls. As monitoring systems become more reliable, the risk at which such accidents have occurred has been going down. But it would be very uncertain to think that technological malfunctions have become impossible. Future technological changes will likely raise the challenges for nuclear weapons control. There may be pressure to integrate artificial intelligence systems into nuclear command and control to allow for faster data processing and decision making. And AI systems are known to behave unexpectedly when deployed in new environments. New technologies will also create new accident risks of their own, even if they're not connected to nuclear weapons systems. Although these risks are hard to predict, they seem significant. I'd say more about how such technologies – including AI, nuclear, biological, and autonomous weapons – are likely to increase war risks later. Leaders could choose war. All that said, most wars have not started by accident. If another great power war does break out in the coming decades, it is more likely to be an intentional decision made by a national leader. Explaining why someone might make such a costly, destructive, unpredictable, and risky decision has been called "the central puzzle about war." It has motivated researchers to search for "rationalist" explanations for war. In his 2022 book Why We Fight, for example, economist Chris Blattman proposes five basic explanations: unbalanced interests, intangible incentives, uncertainty, commitment problems, and misperceptions. This section discusses how great power tensions may escalate to war in the next few decades. It focuses on potential conflicts in particular: war between the US and China, between the US and Russia, and between China and India. These are discussed because each of these countries are among the world's largest economies and military spenders, and were particularly likely to fight. At the end, I briefly touch on other potential large conflicts. United States-China. The most worrying possibility for war is between the United States and China. They are easily the world's largest economies. They spend by far the most on their militaries. Their diplomatic relations are tense and have recently warmed. And their relationship has several of the characteristics that Blattman identifies as causes of war. At the core of the United States-China relationship is a commitment problem. China's economy is growing faster than the United States'. By some metrics, it's already larger 20% if differential growth continues, the gap will continue to widen between it and the United States. While economic power is not the sole determinant of military power, it is a key factor. The United States and China may be able to strike a fair deal today. But as China continues to grow faster, that deal may become unbalanced. Historically, such commitment problems seem to have made these kinds of transition periods particularly dangerous. In practice, the United States and China may find it hard to agree on rules to guide their interactions, such as how to run international institutions or govern areas of the world where their interests overlap. The most obvious issue which could tip the United States-China relationship from tension into war is a conflict over Taiwan. Taiwan's location and technology industries are valuable for both great powers. This issue is further complicated by intangible incentives. For the United States, it is also a conflict about democratic ideals and the United States' reputation for defending its allies. For China, it is also a conflict about territorial integrity and addressing what are seen as past injustices. Still, forecasts suggest that while a conflict is certainly possible, it is far from inevitable. As of 8 June 2021, one aggregated forecast gives a 17% chance of a United States-China war breaking out before 2035. A related aggregated forecast of the chance that at least 100 deaths occur in conflict between China and Taiwan by 2050 gives it, as of 8 June 2021, a much higher 68% chance of occurring. India-Russia is the United States' other major geopolitical rival. Unlike China, Russia is not a rival in economic terms: even after adjusting for purchasing power, its economy is only about one-fifth the size of the United States'. However, Russia denies a substantial fraction of its economy to its military. Crucially, it has the world's largest nuclear arsenal. And Russian leadership has shown a willingness to project power beyond their country's borders. Top five countries by estimated military spending, 2021. Source: SIPRI. Russia's 2022 invasion of Ukraine demonstrates the danger of renewed rivalry between Russia and the United States-led West. The war has already been hugely destructive: the largest war in Europe since World War I, with hundreds of thousands of casualties already and no end to the conflict in sight. And it could get much worse. Most notably, Russian officials have repeatedly refused to rule out the use of nuclear weapons. Unchecked incentives are again at play here. Vladimir Putin leads a highly-centralised government. He has spoken about how his desire to rebuild Russia's reputation played a role in his decision to invade Ukraine. Given their ideological differences and history of rivalry, it is reasonable to expect that the United States and Russia will continue to experience dangerous disagreements in the future. As of 8 June 2021, an aggregated forecast gives a 20% chance that the United States and Russia will fight a war involving at least 1,000 battle deaths before 2050. China-India. India is already the world's third largest economy. If national growth rates remain roughly constant, the size of the Indian economy will surpass that of the United States' sometime this century. India also has nuclear weapons and is already the world's longest military spender (about at a much lower level than China or the United States). One reason to worry that China and India could fight a war is that they already dispute territory along their border. Countries that share a border, especially when it is disputed, are more likely to go to war than countries that do not. By one count, 88% of the wars that occurred between 1816 and 1980 began as wars between neighbours. In fact, China and India already fought a brief but violent border war in 1962. Deadly skirmishes have continued since, resulting in deaths as recently as 2020. Forecasts agree that a China-India conflict seems (relatively though not absolutely) likely. An aggregated forecast gives a 19% chance of a war before 2035. Other dangerous conflicts. These three conflicts – United States-China, United States-Russia, and China-India – are not the only possible great power wars that could occur. Other potential conflicts could also pose existential risk, either because they drive dangerous arms races or use widespread deployment of dangerous weapons. We should keep in mind India-Pakistan as a particularly likely conflict between nuclear-armed states and China-Russia as a potential, though unlikely, conflict between great powers with a disputed border and history of war. Plus, new great powers may emerge or current great powers may fade in the years to come. While I think we should prioritise the three potential conflicts I've highlighted above, the future is highly uncertain. We should monitor geopolitical changes and be open to changing our priorities in the future. Overall predictions. Below is a table listing relevant predictions from the forecasting platform Metaculus, including the number of predictions made, as of 10 March 2023. Note the different likelihoods and resolution criteria for each question; they may not be intuitively comparable. I have previously independently estimated the likelihood of seeing a World War II-like conflict this century. My calculation first adjusts historical base rates to allow for the possibility that major wars have become somewhat less likely, and uses the adjusted base rate to calculate the probability of seeing a war between now and 2100. This method gives a 45% chance of seeing a major great power war in the next 77 years. If the probability is constant over time then the cumulative probability between now and 2050 would be 22%. This is aligned with the Metaculus predictions above. We can also ask experts what they think. Unfortunately, there are surprisingly few expert predictions about the likelihood of major conflict. One survey was conducted by the Project for the Study of the 21st Century. The numbers were relatively aligned with the Metaculus forecasts, though slightly more pessimistic. However, it seems a mistake to put too much stock in this survey (see footnote 27). There may have been at least a rough sense of a great power war's probability. But how bad could it get if it occurred? A new great power war could be devastating. At the time, the mechanised slaughter of World War I was a shocking day-change in the potential severity of warfare. But its severity was surpassed just 20 years later by the outbreak of World War II, which killed over twice as many people. A modern great power war could be even worse. How bad have wars been in the past? The graph below shows how common wars of various sizes are, according to the Correlates of War/Interstate War dataset. The x-axis here represents war size in terms of the logarithm of the number of battle deaths. The y-axis represents the logarithm of the proportion of wars in the dataset that are at least that large. Using logarithms means that each step to the right in the graph represents a war that is not one unit more likely, but 10 times larger. And each step up represents a war that is not one unit more likely, but 10 times more likely. What the graph shows is that wars have a heavy tail. Most wars remain relatively small. But a few wars are greatly and become much worse than average. Of the 95 wars in the latest version of the database, the median battle death toll is 8,000. But the heavy tail means the average is 334,000 battle deaths. The number of battle deaths is only one way to measure the badness of war. We could also consider the proportion of the population of the countries involved who were killed in battle. By this measure, the worst war since 1816 was not World War I. Instead, it's the Paraguayan War of 1864-70. In that war, 30 soldiers died for every 1,000 citizens of the countries involved. It's even worse if we also consider civilian deaths, while estimates are very uncertain, it's plausible that about half of the men in Paraguay, or around a quarter of the entire population, were killed. So what if instead we compared wars by the proportion of the global population killed? World War I is again the worst conflict since 1816 on this measure, having killed about 3% of the global population. Going further back in time, though, we can find worse wars. Ghengis Khan's conquests likely killed about 5.5% of people in the world at the time. The heavy tail means that some wars will be shockingly large. The scale of World War I and World War II took people by surprise, including the leaders who initiated it. It's also hard to know exactly how big wars could get. We haven't seen many really large wars. So while we know there's a heavy tail of potential outcomes, we don't know what that tail looks like. That said, there are a few reasons to think that wars much worse than World War I are possible. We're statistically unlikely to have broken up against the end of the tail, even if the tail has an upper bound. Other wars have been deadlier on a per-capita basis. So unless wars involving countries with larger populations are systematically less intense, we should expect to see more

intense wars involving as many people as World War I. Economic growth and technological progress are continually increasing humanity's war-making capacity. This means that, once a war has started, we're at greater risk of extremely bad outcomes than we were in the past. So how bad could it get?

How bad could a modern great power war be? Over time, two related factors have greatly increased humanity's capacity to make war. First, scientific progress has led to the invention of more powerful weapons and improved military efficiency. Second, economic growth has allowed states to build larger armies and arsenals. Since World War II, the world economy has grown by a factor of more than 10 in real terms; the number of nuclear weapons in the world has grown from basically none to more than 9,000, and we've invented drones, missiles, satellites, and advanced planes, ships, and submarines. Ghengis Khan's conquests killed about 10% of the world, but this took place over the course of two decades. Today that proportion may be killed in a matter of hours. First, **nuclear weapons could be used**. Today there are around 10,000 nuclear warheads globally. At the peak of nuclear competition between the United States and the USSR, though, there were 64,000. If arms control agreements break down and competition resurges among two or even three great powers, nuclear arsenals could expand. In fact, China's arsenal is very likely to grow – though by how much remains uncertain. Many of the nuclear weapons in the arsenals of the great powers today are at least 10 times more powerful than the atomic bombs used in World War II. Should these weapons be used, the consequences would be catastrophic. By any measure, such a war would be by far the most destructive, dangerous event in human history, with the potential to cause billions of deaths. The probability that it would, on its own, lead to humanity's extinction or unrecoverable collapse, is contested. But there seems to be some possibility – whether through a famine caused by nuclear winter, or by reducing humanity's resilience enough that something else, like a catastrophic pandemic, would be far more likely to reach extinction-levels (read more in our problem profile on nuclear war). Nuclear weapons are complemented and amplified by a variety of other modern military technologies, including improved missiles, planes, submarines, and satellites. They are also not the only military technology with the potential to cause a global catastrophe – bioweapons, too, have the potential to cause massive harm through accidents or unexpected effects. What's more, humanity's war-making capacity seems poised to further

increase in the coming years due to technological advances and economic growth. Technological progress could make it cheaper and easier for more states to develop weapons of mass destruction. In some cases, political and economic barriers will remain significant. Nuclear weapons are very expensive to develop and there exists a strong international taboo against their proliferation. In other cases, though, the hurdles to developing extremely powerful weapons may prove lower. Improvements in biotechnology will probably make it cheaper to develop bioweapons. Such weapons may provide the deterrent effect of nuclear weapons at a much lower price. They also seem harder to monitor from abroad, making it more difficult to limit their proliferation. And they could spark a global biological catastrophe, like a major — possibly existentially catastrophic — pandemic. Artificial intelligence systems are also likely to become cheaper as well as more powerful. It is not hard to imagine important military implications of this technology. For example, AI systems could control large groups of lethal autonomous weapons (though the timeline on which such applications will be developed is unclear). They may increase the pace at which war is waged, enabling rapid escalation outside human control. And AI systems could speed up the development of other dangerous new technologies. Finally, we may have to deal with the invention of other weapons which we can't currently predict. The feasibility and danger of nuclear weapons was unclear to many military strategists and scientists until they were first tested. We could similarly experience the invention of destabilising new weapons in our lifetime. What these technologies have in common is the potential to quickly kill huge numbers of people: A nuclear war could kill tens of millions within hours, and many more in the following days and months. A runaway bioweapon could prove very difficult to stop. Future autonomous systems could act with lightning speed, even taking humans out of the decision-making loop entirely. Faster wars leave less time for humans to intervene, negotiate, and find a resolution that limits the damage. How likely is war to damage the long-run future? When a war begins, leaders often promise a quick, limited conflict. But escalation proves hard to predict ahead of time (perhaps because people are scope-insensitive, or because escalation depends on idiosyncratic decisions). This raises the possibility of enormous wars that threaten all of humanity.